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Measure and Integral: An Introduction to Real Analysis, Second Edition

1. Preliminaries

1. Prove the following facts, which were left earlier as exercises.

(a) For a sequence of sets $\{E_k\}$, $\limsup E_k$ consists of those points that belong to infinitely many E_k, and $\liminf E_k$ consists of those points that belong to all E_k from some k on.

(b) The De Morgan laws.

(c) Every Cauchy sequence in $\mathbb{R}^n$ converges to a point of $\mathbb{R}^n$. (This can be deduced from its analogue in $\mathbb{R}^1$ by noting that the entries in a given coordinate position of the points in a Cauchy sequence in $\mathbb{R}^n$ form a Cauchy sequence in $\mathbb{R}^1$.)

(d) Theorem 1.4.

(e) A sequence $\{a_k\}$ in $\mathbb{R}^1$ converges to a, $-\infty \le a \le +\infty$, if and only if $\limsup_{k\to\infty} a_k = \liminf_{k\to\infty} a_k = a$.

(f) $B(x;\delta)$ is open.

(g) Theorem 1.5.

(h) Theorems 1.6 and 1.7.

(i) Theorem 1.8.

(j) Any closed (open) set in $\mathbb{R}^n$ is of type G_δ (F_σ). (If F is closed, consider the sets $\{x : \operatorname{dist}(x, F) < 1/k\}$, $k = 1, 2, \ldots$.)

(k) Theorem 1.12.

(l) The distance between two nonempty, compact, disjoint sets in $\mathbb{R}^n$ is positive. See also Exercise 12.

(m) The intersection of a countable sequence of decreasing, nonempty, compact sets is nonempty.

(n) Theorem 1.14.

(o) Theorem 1.15.

(p) Theorem 1.16.

(q) Theorem 1.17.

(r) The Riemann integral $A = (R)\int_I f(x)\,dx$ of a bounded f over an interval I exists if and only if $\inf_\Gamma U_\Gamma = \sup_\Gamma L_\Gamma = A$.

(s) If f is continuous on an interval I, then $(R)\int_I f(x)\,dx$ exists.

2. Find $\limsup E_k$ and $\liminf E_k$ if $E_k = [-1/k, 1]$ for k odd and $E_k = [-1, 1/k]$ for k even.

3.

(a) Show that $C(\limsup E_k) = \liminf CE_k$.

(b) Show that if $E_k \nearrow E$ or $E_k \searrow E$, then $\limsup E_k = \liminf E_k = E$.

4.

(a) Show that $\limsup_{k\to\infty}(-a_k) = -\liminf_{k\to\infty} a_k$.

(b) Show that $\limsup_{k\to\infty}(a_k+b_k) \le \limsup_{k\to\infty} a_k + \limsup_{k\to\infty} b_k$, provided that the expression on the right does not have the form $\infty + (-\infty)$ or $-\infty + \infty$.

(c) If $\{a_k\}$ and $\{b_k\}$ are nonnegative, bounded sequences, show that $\limsup_{k\to\infty}(a_k b_k) \le (\limsup_{k\to\infty} a_k)(\limsup_{k\to\infty} b_k)$.

(d) Give examples for which the inequalities in parts (b) and (c) are not equalities. Show that if either $\{a_k\}$ or $\{b_k\}$ converges, equality holds in (b) and (c).

5. Find analogues of the statements in Exercise 4 for $\limsup_{x \to x_0;\, x \in E} f(x)$.

6. Compare $\limsup_{k\to\infty} a_k$ and $\limsup(-\infty, a_k)$.

7. Show that $E_1^\circ \cap E_2^\circ = (E_1 \cap E_2)^\circ$ and $E_1^\circ \cup E_2^\circ \subset (E_1 \cup E_2)^\circ$. Give an example when $E_1^\circ \cup E_2^\circ \neq (E_1 \cup E_2)^\circ$.

8. Let E be a set in $\mathbb{R}^n$ that is relatively open with respect to an interval I. Show that E can be written as a countable union of nonoverlapping intervals.

9. Prove that any closed subset of a compact set is compact.

10. Let $\{x_k\}$ be a bounded infinite sequence in $\mathbb{R}^n$. Show that $\{x_k\}$ has a limit point. (This is the Bolzano–Weierstrass theorem in $\mathbb{R}^n$.)

11. Give an example of a decreasing sequence of nonempty closed sets in $\mathbb{R}^n$ whose intersection is empty.

12.

(a) Give an example of two disjoint, nonempty, closed sets E_1 and E_2 in $\mathbb{R}^n$ for which $d(E_1, E_2) = 0$.

(b) Let E_1, E_2 be nonempty sets in $\mathbb{R}^n$ with E_1 closed and E_2 compact. Show that there are points $x_1 \in E_1$ and $x_2 \in E_2$ such that $d(E_1, E_2) = |x_1 - x_2|$. Deduce that $d(E_1, E_2)$ is positive if such E_1, E_2 are disjoint.

13. If f is defined and uniformly continuous on E, show there is a function $\bar{f}$ defined and continuous on $\bar{E}$ such that $\bar{f} = f$ on E.

14. If f is defined and uniformly continuous on a bounded set E, show that f is bounded on E.

15. Show that a bounded f is Riemann integrable on I if and only if given $\varepsilon > 0$, there is a partition Γ of I such that $0 \leq U_\Gamma - L_\Gamma < \varepsilon$. (Exercise 1(r) may be helpful.)

16. If $\{f_k\}$ is a sequence of bounded, Riemann integrable functions on an interval I that converges uniformly on I to f, show that f is Riemann integrable on I and that

$$(R)\int_I f_k(x)\,dx \longrightarrow (R)\int_I f(x)\,dx.$$

17. Let f be a finite function on $\mathbb{R}^n$ and define

$$\omega(\delta) = \sup\{|f(x) - f(y)| : |x - y| \leq \delta\},$$

$\delta > 0$, to be the modulus of continuity of f. Show that $\omega(\delta)$ decreases as δ decreases to 0 and that f is uniformly continuous if and only if $\omega(\delta) \to 0$ as $\delta \to 0$.

18. Let F be a closed subset of $(-\infty, +\infty)$, and let f be continuous relative to F. Show that there is a continuous function g on $(-\infty, +\infty)$ which equals f in F. If $|f(x)| \leq M$ for $x \in F$, show that g can be chosen so that $|g(x)| \leq M$ for $-\infty < x < +\infty$. (This is the Tietze extension theorem for the real line.)

19. Prove the following special case of the Baire category theorem: the intersection of a countable number of open dense sets in $\mathbb{R}^1$ is dense in $\mathbb{R}^1$.

20. Show that the irrational numbers form a set of type G_δ, but that the rational numbers do not. (For the second part, it is possible to argue by contradiction, using the first part and the result in Exercise 19.)

21. Construct a set in $\mathbb{R}^1$ that is neither of type G_δ nor of type F_σ. (Consider the union of the negative rationals and the positive irrationals, and use facts from Exercise 20.)

22. For an integer $k = 1, \ldots, n$ and a real number α, consider the hyperplane $H = \{x = (x_1, \ldots, x_n) : x_k = \alpha\}$. Show that for every $\varepsilon > 0$, there is a collection $\{Q_j\}_{j=1}^{\infty}$ of cubes in $\mathbb{R}^n$ with edges parallel to the coordinate axes such that $H \subset \bigcup Q_j$ and $\sum v(Q_j) < \varepsilon$. (Using the terminology of Chapter 3, it follows that H has outer measure 0 in $\mathbb{R}^n$.)

2. Functions of Bounded Variation and the Riemann–Stieltjes Integral

1. Let $f(x) = x\sin(1/x)$ for $0 < x \leq 1$ and $f(0) = 0$. Show that f is bounded and continuous on $[0, 1]$, but that $V[f; 0, 1] = +\infty$.

2. Prove Theorem 2.1.

3. If $[a', b']$ is a subinterval of $[a, b]$, show that $P[a', b'] \leq P[a, b]$ and $N[a', b'] \leq N[a, b]$.

4. Let $\{f_k\}$ be a sequence of functions of bounded variation on $[a, b]$. If $V[f_k; a, b] \leq M < +\infty$ for all k and if $f_k \to f$ pointwise on $[a, b]$, show that f is of bounded variation and that $V[f; a, b] \leq M$. Give an example of a convergent sequence of functions of bounded variation whose limit is not of bounded variation.

5. Suppose f is finite on $[a, b]$ and of bounded variation on every interval $[a + \varepsilon, b]$, $\varepsilon > 0$, with $V[f; a + \varepsilon, b] \leq M < +\infty$. Show that $V[f; a, b] < +\infty$. Is $V[f; a, b] \leq M$? If not, what additional assumption will make it so?

6. Let $f(x) = x^2 \sin(1/x)$ for $0 < x \leq 1$ and $f(0) = 0$. Show that $V[f; 0, 1] < +\infty$. (Examine the graph of f, or use Exercise 5 and Corollary 2.10.)

7. Suppose f is of bounded variation on $[a, b]$. If f is continuous at a point $\bar{x}$, show that $V(x), P(x)$, and $N(x)$ are also continuous at $\bar{x}$. In particular, if f is continuous on $[a, b]$, then so are $V(x), P(x)$, and $N(x)$. (If $\Gamma = \{x_i\}$, note that $V[x_{i-1}, x_i] - |f(x_{i-1}) - f(x_i)| \leq V[a, b] - S_\Gamma$. Recall that S_Γ increases when Γ is refined.)

8. The main results about functions of bounded variation on a closed bounded interval remain true for open or partly open intervals and for infinite intervals. Prove, for example, that if f is of bounded variation on $(-\infty, +\infty)$, then f is the difference of two increasing bounded functions.

9. Let C be a curve with parametric equations $x = \phi(t)$ and $y = \psi(t)$, $a \le t \le b$.

(a) If ϕ and ψ are of bounded variation and continuous, show that $L = \lim_{|\Gamma|\to 0} l(\Gamma)$.

(b) If ϕ and ψ are continuously differentiable, show that $L = \int_a^b ([\phi'(t)]^2 + [\psi'(t)]^2)^{1/2}\, dt$.

10. If $\lambda_1 < \lambda_2 < \cdots < \lambda_m$ is a finite sequence and $-\infty < s < +\infty$, write $\sum_k a_k e^{-s\lambda_k}$ as a Riemann–Stieltjes integral. (Take $f(x) = e^{-sx}$, ϕ to be an appropriate step function and $[a, b]$ to contain all the λ_k in its interior.)

11. Show that $\int_a^b f\,d\phi$ exists if and only if given $\varepsilon > 0$, there exists $\delta > 0$ such that $|R_\Gamma - R_{\Gamma'}| < \varepsilon$ if $|\Gamma|, |\Gamma'| < \delta$.

12. Prove that the conclusion of Theorem 2.30 is valid if the assumption that ϕ is continuous is replaced by the assumption that f and ϕ have no common discontinuities. (Instead of the uniform continuity of ϕ, use the fact that either f or ϕ is continuous at each point $\bar{x}_j$ of $\bar{\Gamma}$.)

13. Prove Theorem 2.16.

14. Give an example that shows that for $a < c < b$, $\int_a^c f\,d\phi$ and $\int_c^b f\,d\phi$ may both exist but $\int_a^b f\,d\phi$ may not. Compare Theorem 2.17. (Take $[a, b] = [-1, 1]$, $c = 0$, and f and ϕ as in the example following (2.28).)

15. Suppose f is continuous and ϕ is of bounded variation on $[a, b]$. Show that the function $\psi(x) = \int_a^x f\, d\phi$ is of bounded variation on $[a, b]$. If g is continuous on $[a, b]$, show that $\int_a^b g\, d\psi = \int_a^b gf\, d\phi$.

16. Suppose that ϕ is of bounded variation on $[a, b]$ and that f is bounded and continuous except for a finite number of jump discontinuities in $[a, b]$. If ϕ is continuous at each discontinuity of f, show that $\int_a^b f\, d\phi$ exists.

17. If ϕ is of bounded variation on $(-\infty, +\infty)$, f is continuous on $(-\infty, +\infty)$, and $\lim_{|x|\to+\infty} f(x) = 0$, show that $\int_{-\infty}^{+\infty} f\, d\phi$ exists.

18. Let $f(z) = \sum_{k=0}^{\infty} a_k z^k$ be a power series. Show that if $\sum |a_k| < +\infty$, then $f(z)$ is of bounded variation on every radius of the circle $|z| = 1$. (If, e.g., the radius is $0 \le x \le 1$ and the a_k are real, then $f(x) = \sum a_k^+ x^k - \sum a_k^- x^k$.)

19. Let f and ϕ be finite functions on $[a, b]$. If $[a', b'] \subset [a, b]$ and $\int_a^b f\, d\phi$ exists, show that $\int_{a'}^{b'} f\, d\phi$ exists.

20. Let f be finite on $[a, b]$. If $\lim_{|\Gamma|\to 0} S_\Gamma[f; a, b]$ exists, show that it equals $V[f; a, b]$.

21. If $V[\phi; a, b] = +\infty$, show that there is a point $x_0 \in [a, b]$ such that either $V[\phi; I] = +\infty$ for every subinterval I of $[a, b]$ having x_0 as left-hand endpoint or $V[\phi; I] = +\infty$ for every subinterval I of $[a, b]$ having x_0 as right-hand endpoint. (Note that for any x_0, a', b' such that $a \leq a' < x_0 < b' \leq b$, $V[\phi; a', b'] < +\infty$ if both $V[\phi; a', x_0], V[\phi; x_0, b'] < +\infty$.)

22. If $V[\phi; a, b] = +\infty$, show that there exist $x_0 \in [a, b]$ and a monotone sequence $\{x_k\}_1^\infty$ in $[a, b]$ such that $x_k \to x_0$ and $\sum_{k=1}^{\infty} |\phi(x_{k+1}) - \phi(x_k)| = +\infty$. (Use the results in Exercises 21 and 5.)

23. If $V[\phi; a, b] = +\infty$, show that there is a continuous f on $[a, b]$ such that $\int_a^b f\, d\phi$ does not exist. (Use the result in Exercise 22 together with the following fact: if $\{\alpha_k\}$ is a sequence of positive numbers with $\sum \alpha_k = +\infty$, then there is a sequence $\{\varepsilon_k\}$ of positive numbers with $\varepsilon_k \to 0$ and $\sum \varepsilon_k \alpha_k = +\infty$.)

24. Let f be continuous and ϕ be of bounded variation on $[a, b]$, and recall that the Riemann–Stieltjes integral $\int_a^b f\, d\phi$ then exists by Theorem 2.24. Show that $\lim_{\varepsilon \to 0+} \int_a^{a+\varepsilon} f\, d\phi = 0$ if and only if either $f(a) = 0$ or ϕ is continuous at a. Deduce that the formula $\int_a^b f\, d\phi = \lim_{\varepsilon \to 0+} \int_{a+\varepsilon}^b f\, d\phi$ may not hold.

25. Construct a bounded nondecreasing function on $(-\infty, \infty)$ which is continuous at every irrational number and discontinuous at every rational number. (Let $\{r_k\}_{k=1}^{\infty}$ be an enumeration of the rational numbers and consider the function $f(x) = \sum_{k:r_k \leq x} 2^{-k}$.)

26. Let $f(x) = \sin x$. Sketch the graphs of its variations $P(x), N(x), V(x)$ for $0 \leq x \leq 2\pi$, and use Corollary 2.10 to find explicit formulas for these variations on $[0, 2\pi]$.

27. If f is an even function on $[-1, 1]$, verify the formulas $V[f; -1, 1] = 2P[f; -1, 1]$, $V[f; 0, 1] = P[f; -1, 1]$ and $P[f; 0, 1] = N[f; -1, 0]$.

28. Let f be Lipschitz continuous on an interval $[a, b]$. Show that $V[f; a, b]$ is strictly less than the length of the graph of f over $[a, b]$.

29. Use Theorem 2.17 to verify the formula for $\int_a^b f\, d\phi$ given in remark 3 in Section 2.3.

30. Let f and ϕ be real-valued functions on $[a, b]$.

(a) If $\int_a^b f\, d\phi$ exists and ϕ is not constant on any subinterval of $[a, b]$, show that f is bounded on $[a, b]$.

(b) If $\int_a^b f\, d\phi$ exists and ϕ is increasing, show that $L_\Gamma \le \int_a^b f\, d\phi \le U_\Gamma$ for every partition Γ of $[a, b]$, where L_Γ and U_Γ are the corresponding lower and upper sums.

31. Show that for any real-valued function f on $[a, b]$, the Riemann–Stieltjes integral $\int_a^b df$ exists and equals $f(b)-f(a)$. If f' exists and is Riemann integrable on $[a, b]$, show that $\int_a^b f'(x)\,dx = \int_a^b df$, and consequently $\int_a^b f'(x)\,dx = f(b) - f(a)$.

32. Let f be a function of bounded variation on an interval $[a, b]$. Show that f is Riemann integrable on $[a, b]$. (This follows from Theorems 5.54 and 2.8, but it can be derived solely from the results in this chapter.)

3. Lebesgue Measure and Outer Measure

1. There is an analogue for bases different from 10 of the usual decimal expansion of a number. If b is an integer larger than 1 and $0 \leq x \leq 1$, show that there exist integral coefficients c_k, $0 \leq c_k < b$, such that $x = \sum_{k=1}^{\infty} c_k b^{-k}$. Show that this expansion is unique unless $x = cb^{-k}$, $c = 1, \ldots, b^k - 1$, in which case there are two expansions.

2. When $b = 3$ in Exercise 1, the expansion is called the triadic or ternary expansion of x.

(a) Show that the Cantor set C consists of all x such that x has some triadic expansion for which every c_k is either 0 or 2.

(b) Let $f(x)$ be the Cantor–Lebesgue function: see p. 43 in Section 3.1. Show that if $x \in C$ and $x = \sum c_k 3^{-k}$, where each c_k is either 0 or 2, then $f(x) = \sum(\frac{1}{2}c_k)2^{-k}$.

3. Construct a two-dimensional Cantor set in the unit square $\{(x, y) : 0 \leq x, y \leq 1\}$ as follows. Subdivide the square into nine equal parts and keep only the four closed corner squares, removing the remaining region (which forms a cross). Then repeat this process in a suitably scaled version for the remaining squares, ad infinitum. Show that the resulting set is perfect, has plane measure zero, and equals $C \times C$.

4. Construct a subset of $[0, 1]$ in the same manner as the Cantor set by removing from each remaining interval a subinterval of relative length θ, $0 < \theta < 1$. Show that the resulting set is perfect and has measure zero.

5. Construct a subset of $[0,1]$ in the same manner as the Cantor set, except that at the kth stage, each interval removed has length $\delta 3^{-k}$, $0 < \delta < 1$. Show that the resulting set is perfect, has measure $1 - \delta$, and contains no intervals.

6. Construct a Cantor-type subset of $[0,1]$ by removing from each interval remaining at the kth stage a subinterval of relative length θ_k, $0 < \theta_k < 1$. Show that the remainder has measure zero if and only if $\sum \theta_k = +\infty$. (Use the fact that for $a_k > 0$, $\prod_{k=1}^{\infty} a_k$ converges, in the sense that $\lim_{N\to\infty} \prod_{k=1}^{N} a_k$ is finite and not zero, if and only if $\sum_{k=1}^{\infty} \log a_k$ converges.)

7. Prove Lemma 3.15.

8. Show that the Borel σ-algebra $\mathcal{B}$ in $\mathbb{R}^n$ is the smallest σ-algebra containing the closed sets in $\mathbb{R}^n$.

9. If $\{E_k\}_{k=1}^{\infty}$ is a sequence of sets with $\sum |E_k|_e < +\infty$, show that $\limsup E_k$ (and so also $\liminf E_k$) has measure zero.

10. If E_1 and E_2 are measurable, show that $|E_1 \cup E_2| + |E_1 \cap E_2| = |E_1| + |E_2|$.

11. Prove Theorem 3.29. (For the sufficiency, pick open sets G and G_1 with $S \subset G$, $N_1 \subset G_1$, $|G - S| < \varepsilon$, and $|G_1| < |N_1|_e + \varepsilon < 2\varepsilon$. Estimate $|(G \cup G_1) - E|_e$.)

12. If E_1 and E_2 are measurable subsets of $\mathbb{R}^1$, show that $E_1 \times E_2$ is a measurable subset of $\mathbb{R}^2$ and $|E_1 \times E_2| = |E_1||E_2|$. (Interpret $0 \cdot \infty$ as 0.) (Hint: Use a characterization of measurability.)

13. Motivated by (3.7), define the inner measure of E by $|E|_i = \sup |F|$, where the supremum is taken over all closed subsets F of E. Show that (i) $|E|_i \leq |E|_e$, and (ii) if $|E|_e < +\infty$, then E is measurable if and only if $|E|_i = |E|_e$. (Use Lemma 3.22.)

14. Show that the conclusion of part (ii) of Exercise 13 is false if $|E|_e = +\infty$.

15. If E is measurable and A is any subset of E, show that $|E| = |A|_i + |E - A|_e$. (See Exercise 13 for the definition of $|A|_i$.) As a consequence, using Exercise 13, show that if $A \subset [0,1]$ and $|A|_e + |[0,1] - A|_e = 1$, then A is measurable.

16. Prove (3.34).

17. Give an example that shows that the image of a measurable set under a continuous transformation may not be measurable. (Consider the Cantor–Lebesgue function and the pre-image of an appropriate nonmeasurable subset of its range.) See also Exercise 10 of Chapter 7.

18. Prove that outer measure is translation invariant; that is, if $E_h = \{x + h : x \in E\}$ is the translate of E by h, $h \in \mathbb{R}^n$, show that $|E_h|_e = |E|_e$. If E is measurable, show that E_h is also measurable. (This fact was used in proving Lemma 3.37.)

19. Carry out the details of the construction of a nonmeasurable subset of $\mathbb{R}^n$, $n > 1$.

20. Show that there exist disjoint $E_1, E_2, \ldots, E_k, \ldots$ such that $|\bigcup E_k|_e < \sum |E_k|_e$ with strict inequality. (Let E be a nonmeasurable subset of $[0,1]$ whose rational translates are disjoint. Consider the translates of E by all rational numbers r, $0 < r < 1$, and use Exercise 18.)

21. Show that there exist sets $E_1, E_2, \ldots, E_k, \ldots$ such that $E_k \searrow E$, $|E_k|_e < +\infty$, and $\lim_{k\to\infty} |E_k|_e > |E|_e$ with strict inequality.

22.

(a) Show that the outer measure of a set is unchanged if in the definition of outer measure we use coverings of the set by cubes with edges parallel to the coordinate axes instead of coverings by intervals.

(b) Show that outer measure is also unchanged if coverings by parallelepipeds with a fixed orientation (i.e., with edges parallel to a fixed set of n linearly independent vectors) are used rather than coverings by intervals.

23. Let Z be a subset of $\mathbb{R}^1$ with measure zero. Show that the set $\{x^2 : x \in Z\}$ also has measure zero.

24. Let $0.\alpha_1\alpha_2\ldots$ be the dyadic development of any x in $[0,1]$, that is, $x = \alpha_1 2^{-1} + \alpha_2 2^{-2} + \cdots$ with $\alpha_i = 0, 1$. Let $k_1, k_2, \ldots$ be a fixed permutation of the positive integers $1, 2, \ldots$, and consider the transformation T that sends $x = 0.\alpha_1\alpha_2\cdots$ to $Tx = 0.\alpha_{k_1}\alpha_{k_2}\ldots$. If E is a measurable subset of $[0,1]$, show that its image TE is also measurable and that $|TE| = |E|$. (Consider first the special cases of E a dyadic interval $[s2^{-k}, (s+1)2^{-k}]$, $s = 0, 1, \ldots, 2^k - 1$, and then of E an open set [which is a countable union of nonoverlapping dyadic intervals]. Also show that if E has small measure, then so has TE.)

25. Construct a measurable subset E of $[0,1]$ such that for every subinterval I, both $E \cap I$ and $I - E$ have positive measure. (Take a Cantor-type subset of $[0,1]$ with positive measure [see Exercise 5], and on each subinterval of the complement of this set, construct another such set, and so on. The measures can be arranged so that the union of all the sets has the desired property.) See also Exercise 21 of Chapter 4.

26. Construct a continuous function on $[0, 1]$, which is not of bounded variation on any subinterval. (The construction follows the pattern of the Cantor–Lebesgue function with some modifications. At the first stage, for example, make the corresponding function increase to $2/3$ [rather than $1/2$] in $(0, 1/3)$, then make it decrease by $1/3$ in $(1/3, 2/3)$, and then increase again $2/3$ in $(2/3, 1)$. The construction at other stages is similar, depending on whether the preceding function was increasing or decreasing in the subinterval under consideration.)

27. Construct a continuous function of bounded variation on $[0, 1]$ which is not monotone in any subinterval. (The construction is like that in the preceding exercise, except that the approximating functions are less steep. For example, at the first stage, let the function increase to $1/2 + \varepsilon$, then decrease by 2ε, and then increase again by $1/2 + \varepsilon$. Choose the ε's at each stage so that their sum converges.)

28. Prove the following assertion that is made in the proof of Theorem 3.33: If $T : \mathbb{R}^n \to \mathbb{R}^n$ is a Lipschitz transformation, then there is a constant $c' > 0$ such that $|TI| \leq c'|I|$ for every interval I. (Consider first the case when I is a cube Q, noting that TQ is contained in a cube with edge length $c \operatorname{diam} Q$ where c is the Lipschitz constant of T. The case of general I can then be deduced by applying Theorem 1.11 to the interiors J° of intervals J with $I \subset J^\circ$.)

29. Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be a linear transformation.

(a) If T has matrix representation (t_{ij}) and $t = (\sum_{i,j} t_{ij}^2)^{1/2}$, show that $|Tx - Ty| \le t|x - y|$ for all $x, y \in \mathbb{R}^n$. (Use (1.2).) The number t is called the Hilbert–Schmidt norm of (t_{ij}).

(b) Prove the fact mentioned at the end of the proof of Theorem 3.35 that $|TE|_e = \delta|E|_e$ for every $E \subset \mathbb{R}^n$, where $\delta = |\det T|$.

30. Let $f : \mathbb{R}^n \to \mathbb{R}^1$ be continuous. Show that the inverse image $f^{-1}(B)$ of a Borel set B is a Borel set; see p. 64 in Section 4.1 for the definition of the inverse image of a set. (The collection of sets $\{E : f^{-1}(E)$ is a Borel set$\}$ is a σ-algebra and contains all open sets in $\mathbb{R}^1$; cf. Exercise 10 of Chapter 4 and Corollary 4.15.) See also Exercise 22 of Chapter 4.

31. Construct a Lebesgue measurable set that is not Borel measurable. (If f is the Cantor–Lebesgue function, then the function $g(x) = x + f(x)$ is strictly increasing and continuous on $[0, 1]$. Consider the set $g^{-1}(E)$ for an appropriate $E \subset g(C)$, where C is the Cantor set.)

32. Let E be a set in $\mathbb{R}^n$ with $|E|_e > 0$ and let θ satisfy $0 < \theta < 1$. Show that there is a set $E_\theta \subset E$ with $|E_\theta|_e = \theta |E|_e$ and that E_θ can be chosen to be measurable if E is measurable. (If $Q(r)$ denotes the cube with edge length r centered at the origin, $0 < r < \infty$, then $|E \cap Q(r)|_e$ is a continuous monotone function of r.)

33. Let E be a measurable set with $0 < |E| \leq \infty$. Show that there are infinite collections $\{A_j\}, \{B_j\}$ of measurable subsets of E with the following properties: $0 < |A_j|, |B_j| < \infty$, $A_j \cap A_l = \varnothing$ if $j \neq l$, $B_{j+1} \subset B_j$, and $|B_j| \to 0$. See the end of Section 8.5 for an application. (This can be proved in many ways, for example, by using the Exercise 32.)

34. Let E and Z be sets in $\mathbb{R}^n$ and $|Z| = 0$. If $E \cup Z$ is measurable, show that E is measurable and that $|E| = |E \cup Z|$. See Exercise 2 of Chapter 10.

4. Lebesgue Measurable Functions

1. Prove Corollary 4.2 and Theorem 4.8.

2. Let f be a simple function, taking its distinct values on disjoint sets $E_1, \ldots, E_N$. Show that f is measurable if and only if $E_1, \ldots, E_N$ are measurable.

3. Theorem 4.3 can be used to define measurability for vector-valued (e.g., complex-valued) functions. Suppose, for example, that f and g are real-valued and finite in $\mathbb{R}^n$, and let $F(x) = (f(x), g(x))$. Then F is said to be measurable if $F^{-1}(G)$ is measurable for every open $G \subset \mathbb{R}^2$. Prove that F is measurable if and only if both f and g are measurable in $\mathbb{R}^n$.

4. Let f be defined and measurable in $\mathbb{R}^n$. If T is a nonsingular linear transformation of $\mathbb{R}^n$, show that $f(Tx)$ is measurable. (If $E_1 = \{x : f(x) > a\}$ and $E_2 = \{x : f(Tx) > a\}$, show that $E_2 = T^{-1}E_1$.)

5. Give an example to show that $\phi(f(x))$ may not be measurable if ϕ and f are measurable and finite. (Let F be the Cantor–Lebesgue function and let f be its inverse, suitably defined. Let ϕ be the characteristic function of a set of measure zero whose image under F is not measurable.) Show that the same may be true even if f is continuous. (Let $g(x) = x + F(x)$, where F is the Cantor–Lebesgue function, and consider $f = g^{-1}$.) Cf. Exercise 22.

6. Let f and g be measurable functions on E:

(a) If f and g are finite a.e. in E, show that $f+g$ is measurable no matter how we define it at the points when it has the form $+\infty+(-\infty)$ or $-\infty+\infty$.

(b) Show that fg is measurable without restriction on the finiteness of f and g. Show that $f+g$ is measurable if it is defined to have the same value at every point where it has the form $+\infty+(-\infty)$ or $-\infty+\infty$. (Note that a function h defined on E is measurable if and only if both $\{h=+\infty\}$ and $\{h=-\infty\}$ are measurable and the restriction of h to the subset of E where h is finite is measurable.)

7. Let f be usc and less than $+\infty$ on a compact set E. Show that f is bounded above on E. Show also that f assumes its maximum on E, that is, that there exists $x_0 \in E$ such that $f(x_0) \geq f(x)$ for all $x \in E$.

8.

(a) Let f and g be two functions that are usc at x_0. Show that $f+g$ is usc at x_0. Is $f-g$ usc at x_0? When is fg usc at x_0?

(b) If $\{f_k\}$ is a sequence of functions that are usc at x_0, show that $\inf_k f_k(x)$ is usc at x_0.

(c) If $\{f_k\}$ is a sequence of functions that are usc at x_0 and converge uniformly near x_0, show that $\lim f_k$ is usc at x_0.

9.

(a) Show that the limit of a decreasing (increasing) sequence of functions usc (lsc) at x_0 is usc (lsc) at x_0. In particular, the limit of a decreasing (increasing) sequence of functions continuous at x_0 is usc (lsc) at x_0.

(b) Let f be usc and less than $+\infty$ on $[a, b]$. Show that there exist continuous f_k on $[a, b]$ such that $f_k \searrow f$. (First show that there are usc step functions $f_k \searrow f$.)

10.

(a) If f is defined and continuous on E, show that $\{a < f < b\}$ is relatively open and that $\{a \leq f \leq b\}$ and $\{f = a\}$ are relatively closed.

(b) Let f be a finite function on $\mathbb{R}^n$. Show that f is continuous on $\mathbb{R}^n$ if and only if $f^{-1}(G)$ is open for every open G in $\mathbb{R}^1$, or if and only if $f^{-1}(F)$ is closed for every closed F in $\mathbb{R}^1$.

11. Let f be defined on $\mathbb{R}^n$ and let $B(x)$ denote the open ball $\{y : |x - y| < r\}$ with center x and fixed radius r. Show that the function $g(x) = \sup\{f(y) : y \in B(x)\}$ is lsc and that the function $h(x) = \inf\{f(y) : y \in B(x)\}$ is usc on $\mathbb{R}^n$. Is the same true for the closed ball $\{y : |x - y| \leq r\}$?

12. If $f(x)$, $x \in \mathbb{R}^1$, is continuous at almost every point of an interval $[a, b]$, show that f is measurable on $[a, b]$. Generalize this to functions defined in $\mathbb{R}^n$. (For a constructive proof, use the subintervals of a sequence of partitions to define a sequence of simple measurable functions converging to f a.e. in $[a, b]$. Use Theorem 4.12. See also the proof of Theorem 5.54.)

13. One difficulty encountered in trying to extend the proof of Egorov's theorem to the continuous parameter case $f_y(x) \to f(x)$ as $y \to y_0$ is showing that the analogues of the sets E_m in Lemma 4.18 are measurable. This difficulty can often be overcome in individual cases. Suppose, for example, that $f(x, y)$ is defined and continuous in the square $0 \leq x \leq 1$, $0 < y \leq 1$ and that $f(x) = \lim_{y \to 0} f(x, y)$ exists and is finite for x in a measurable subset E of $[0, 1]$. Show that if ε and δ satisfy $0 < \varepsilon, \delta < 1$, the set $E_{\varepsilon,\delta} = \{x \in E : |f(x, y) - f(x)| \leq \varepsilon \text{ for all } y < \delta\}$ is measurable. (If y_k, $k = 1, 2, \ldots$, is a dense subset of $(0, \delta)$, show that $E_{\varepsilon,\delta} = \bigcap_k \{x \in E : |f(x, y_k) - f(x)| \leq \varepsilon\}$.)

14. Let $f(x, y)$ be as in Exercise 13. Show that given $\varepsilon > 0$, there exists a closed $F \subset E$ with $|E - F| < \varepsilon$ such that $f(x, y)$ converges uniformly for $x \in F$ to $f(x)$ as $y \to 0$. (Follow the proof of Egorov's theorem, using the sets $E_{\varepsilon,1/m}$ defined in Exercise 13 in place of the sets E_m in the proof of Lemma 4.18.)

15. Let $\{f_k\}$ be a sequence of measurable functions defined on a measurable E with $|E| < +\infty$. If $|f_k(x)| \leq M_x < +\infty$ for all k for each $x \in E$, show that given $\varepsilon > 0$, there is a closed $F \subset E$ and a finite M such that $|E - F| < \varepsilon$ and $|f_k(x)| \leq M$ for all k and all $x \in F$.

16. Prove that $f_k \xrightarrow{m} f$ on E if and only if given $\varepsilon > 0$, there exists K such that $|\{|f - f_k| > \varepsilon\}| < \varepsilon$ if $k > K$. Give an analogous Cauchy criterion.

17. Suppose that $f_k \xrightarrow{m} f$ and $g_k \xrightarrow{m} g$ on E. Show that $f_k+g_k \xrightarrow{m} f+g$ on E and, if $|E| < +\infty$, that $f_k g_k \xrightarrow{m} fg$ on E. If, in addition, $g_k \to g$ on E, $g \neq 0$ a.e., and $|E| < +\infty$, show that $f_k/g_k \xrightarrow{m} f/g$ on E. (For the product $f_k g_k$, write $f_k g_k - fg = (f_k - f)(g_k - g) + f(g_k - g) + g(f_k - f)$. Consider each term separately, using the fact that a function that is finite on E, $|E| < +\infty$ is bounded outside a subset of E with small measure.)

18. If f is measurable on E, define $\omega_f(a) = |\{f > a\}|$ for $-\infty < a < +\infty$. If $f_k \nearrow f$, show that $\omega_{f_k} \nearrow \omega_f$. If $f_k \xrightarrow{m} f$, show that $\omega_{f_k} \to \omega_f$ at each point of continuity of ω_f. (For the second part, show that if $f_k \xrightarrow{m} f$, then $\limsup_{k\to\infty} \omega_{f_k}(a) \leq \omega_f(a-\varepsilon)$ and $\liminf_{k\to\infty} \omega_{f_k}(a) \geq \omega_f(a+\varepsilon)$ for every $\varepsilon > 0$.)

19. Let $f(x, y)$ be a function defined on the unit square $0 \leq x \leq 1$, $0 \leq y \leq 1$ which is continuous in each variable separately. Show that f is a measurable function of (x, y). Is the same true if f is only assumed to be continuous in x for each y?

20. If f is measurable and finite a.e. on $[a, b]$, show that given $\varepsilon > 0$, there is a continuous g on $[a, b]$ such that $|\{x : f(x) \neq g(x)\}| < \varepsilon$. (See Exercise 18 of Chapter 1.) Formulate and prove a similar result in $\mathbb{R}^n$ by combining Lusin's theorem with the Tietze extension theorem.

21. Show that the necessity part of Lusin's theorem is not true for $\varepsilon = 0$, that is, find a measurable set E and a finite measurable function f on E such that f is not continuous relative to $E - Z$ for any Z with $|Z| = 0$. (Consider, e.g., χ_E for the set E in Exercise 25 of Chapter 3.)

22.

(a) Show that if f is measurable and B is a Borel set in $\mathbb{R}^1$, then $f^{-1}(B)$ is measurable. (Recall that the Borel sets form the smallest σ-algebra that contains the open sets. Consider the collection of sets $\{E : f^{-1}(E)$ is measurable$\}$.) Cf. Exercise 30 of Chapter 3.

(b) If ϕ is a Borel measurable function on $\mathbb{R}^1$ and f is finite and measurable on $\mathbb{R}^n$, show that $\phi(f(x))$ is measurable on $\mathbb{R}^n$. Cf. Theorem 4.6 and Exercise 5.

23. Let $\{f_k\}_{k=1}^{\infty}$ be a sequence of measurable functions defined on a measurable set E. Show that the sets $\{x : \lim f_k(x) \text{ exists and is finite}\}$, $\{x : \lim f_k(x) = +\infty\}$, $\{x : \lim f_k(x) = -\infty\}$ are measurable.

24. Let f be a (Lebesgue) measurable function defined on a (Lebesgue) measurable set E. Show that there is a Borel set $H \subset E$ and a Borel measurable function h on H such that $|E| = |H|$ and $f = h$ on H. (This can be deduced from Lusin's theorem.) In case E is a Borel set, and consequently the exceptional set $Z = E - H$ is also a Borel set, show that f can be redefined on Z so that the resulting function is Borel measurable on E.

5. The Lebesgue Integral

1. If f is a simple measurable function (not necessarily nonnegative) taking values a_j on E_j, $j = 1, 2, \ldots, N$, show that $\int_E f = \sum_{j=1}^{N} a_j |E_j|$. (Use Theorem 5.24.)

2. Show that the conclusions of Theorem 5.32 are not generally true without the assumption that $\phi \in L(E)$. (In part (ii), for example, take $f_k = \chi_{(k,\infty)}$.) Show that Theorem 5.33 fails without the assumption that $|E| < \infty$.

3. Let $\{f_k\}$ be a sequence of nonnegative measurable functions defined on E. If $f_k \to f$ and $f_k \leq f$ a.e. on E, show that $\int_E f_k \to \int_E f$.

4. If $f \in L(0,1)$, show that $x^k f(x) \in L(0,1)$ for $k = 1, 2, \ldots$, and that $\int_0^1 x^k f(x)\,dx \to 0$.

5. Use Egorov's theorem to prove the bounded convergence theorem.

6. Let $f(x, y)$, $0 \le x, y \le 1$, satisfy the following conditions: for each x, $f(x, y)$ is an integrable function of y, and $\partial f(x, y)/\partial x$ is a bounded function of (x, y). Show that $\partial f(x, y)/\partial x$ is a measurable function of y for each x and

$$\frac{d}{dx}\int_0^1 f(x, y)\, dy = \int_0^1 \frac{\partial}{\partial x} f(x, y)\, dy.$$

7. Give an example of an f that is not integrable, but whose improper Riemann integral exists and is finite.

8. Prove (5.49).

9. If $p > 0$ and $\int_E |f - f_k|^p \to 0$ as $k \to \infty$, show that $f_k \xrightarrow{m} f$ on E (and thus that there is a subsequence $f_{k_j} \to f$ a.e. in E).

10. If $p > 0$, $\int_E |f - f_k|^p \to 0$, and $\int_E |f_k|^p \leq M$ for all k, show that $\int_E |f|^p \leq M$.

11. For which $p > 0$ does $1/x \in L^p(0,1)$? $L^p(1,\infty)$? $L^p(0,\infty)$?

12. Give an example of a bounded continuous f on $(0,\infty)$ such that $\lim_{x\to\infty} f(x) = 0$ but $f \notin L^p(0,\infty)$ for any $p > 0$.

13.

(a) Let $\{f_k\}$ be a sequence of measurable functions on E. Show that $\sum f_k$ converges absolutely a.e. in E if $\sum \int_E |f_k| < +\infty$. (Use Theorems 5.16 and 5.22.)

(b) If $\{r_k\}$ denotes the rational numbers in $[0,1]$ and $\{a_k\}$ satisfies $\sum |a_k| < +\infty$, show that $\sum a_k |x - r_k|^{-1/2}$ converges absolutely a.e. in $[0,1]$.

14. Prove the following result (which is obvious if $|E| < +\infty$), describing the behavior of $a^p\omega(a)$ as $a \to 0+$. If $f \in L^p(E)$, then $\lim_{a\to 0+} a^p\omega(a) = 0$. (If $f \geq 0$, $\varepsilon > 0$, choose $\delta > 0$ so that $\int_{\{f\leq\delta\}} f^p < \varepsilon$. Thus, $a^p[\omega(a) - \omega(\delta)] \leq \int_{\{a<f\leq\delta\}} f^p < \varepsilon$ for $0 < a < \delta$. Now let $a \to 0$.)

15. Suppose that f is nonnegative and measurable on E and that ω is finite on $(0, \infty)$. If $\int_0^\infty \alpha^{p-1}\omega(\alpha)\,d\alpha$ is finite, show that $\lim_{a\to 0+} a^p\omega(a) = \lim_{b\to+\infty} b^p\omega(b) = 0$. (Consider $\int_{a/2}^a$ and $\int_{b/2}^b$.)

16. Suppose that f is nonnegative and measurable on E and that ω is finite on $(0, \infty)$. Show that Theorem 5.51 holds without any further restrictions (i.e., f need not be in $L^p(E)$ and $|E|$ need not be finite) if we interpret $\int_0^\infty \alpha^p \, d\omega(\alpha) = \lim_{\substack{a \to 0+ \\ b \to 0+\infty}} \int_b^a$. (For the first part, use the sets E_{ab} to obtain the relation $\int_E f^p = -\int_0^\infty \alpha^p \, d\omega(\alpha)$. If either $\int_0^\infty \alpha^p \, d\omega(\alpha)$ or $\int_0^\infty \alpha^{p-1} \omega(\alpha) \, d\alpha$ is finite, use Lemma 5.50 and the results of Exercises 14 or 15 to integrate by parts.)

17. If $f \geq 0$ and $\omega(\alpha) \leq c(1+\alpha)^{-p}$ for all $\alpha > 0$, show that $f \in L^r$, $0 < r < p$.

18. If $f \geq 0$, show that $f \in L^p$ if and only if $\sum_{k=-\infty}^{+\infty} 2^{kp}\omega(2^k) < +\infty$. (Use Exercise 16.)

19. Derive analogues of Theorems 5.52 and 5.54 for integrals over intervals in $\mathbb{R}^n$, $n > 1$.

20. Let $y = Tx$ be a nonsingular linear transformation of $\mathbb{R}^n$. If $\int_E f(y)\,dy$ exists, show that

$$\int_E f(y)\,dy = |\det T| \int_{T^{-1}E} f(Tx)\,dx.$$

(The case when $f = \chi_{E_1}$, $E_1 \subset E$, follows from integrating the formula $\chi_{E_1}(Tx) = \chi_{T^{-1}E_1}(x)$ over $T^{-1}E$ and then applying Theorem 3.35.)

21. If $\int_A f = 0$ for every measurable subset A of a measurable set E, show that $f = 0$ a.e. in E.

22. Show that the conclusion of the Lebesgue dominated convergence theorem can be strengthened to $\int_E |f_k - f| \to 0$.

23. Prove the following fact, sometimes referred to as the Sequential (or Generalized) Version of the Lebesgue Dominated Convergence Theorem. Let $\{f_k\}$ and $\{\phi_k\}$ be sequences of measurable functions on E satisfying $f_k \to f$ a.e. in E, $\phi_k \to \phi$ a.e. in E, and $|f_k| \leq \phi_k$ a.e. in E. If $\phi \in L(E)$ and $\int_E \phi_k \to \int_E \phi$, then $\int_E |f_k - f| \to 0$. (In case $f = 0$ and all $f_k \geq 0$, apply Fatou's lemma to $\{\phi_k - f_k\}$.) An application is given in Exercise 12 of Chapter 8; for example, if $f_k \geq 0$, $f_k \to f$ a.e. in E, $f \in L(E)$, and $\int_E f_k \to \int_E f$, then $\int_E |f_k - f| \to 0$.

24. A measurable function f on E is said to belong to weak $L^p(E)$, $0 < p < \infty$, if there is a constant $A \geq 0$ such that $\omega_{|f|}(\alpha) \leq A\alpha^{-p}$ for all $\alpha > 0$ (cf. (7.8) in case $p = 1$):

(a) Show that if $f \in L^p(E)$, then f belongs to weak $L^p(E)$, but that the converse is generally false.

(b) Show that if $1 < p < r < \infty$ and f belongs to both weak $L^1(E)$ and weak $L^r(E)$, then $f \in L^p(E)$.

(c) Show that if f belongs to weak $L^1(E)$ and f is bounded on E, then $f \in L^p(E)$ for all $1 < p < \infty$.

25. Give an example to show that the analogue of Theorem 5.8 with the roles of sup and inf interchanged is false.

26. Prove the following variant of Lebesgue's dominated convergence theorem: if $\{f_k\}$ satisfies $f_k \xrightarrow{m} f$ on E and $|f_k| \leq \phi \in L(E)$, then $f \in L(E)$ and $\int_E f_k \to \int_E f$. (Show that every subsequence of $\{f_k\}$ has a subsequence $\{f_{k_j}\}$ such that $\int_E f_{k_j} \to \int_E f$.)

27. The notion of equimeasurability of functions can be extended to different sets E_1 and E_2, even in different dimensions, by saying that two measurable functions f_1, f_2 defined on E_1, E_2, respectively, are equimeasurable if $|\{x \in E_1 : f_1(x) > \alpha\}| = |\{y \in E_2 : f_2(y) > \alpha\}|$ for all α.

(a) Show that if f is measurable and finite a.e. in E and ω_f is strictly decreasing and continuous, then f and the inverse function of ω_f are equimeasurable (on E and $(0, |E|)$, respectively).

(b) Let f be measurable and finite a.e. in E, and suppose that ω_f is finite. Define $f^*(t) = \inf\{\alpha > 0 : \omega_f(\alpha) \leq t\}$, $t > 0$. Show that f and f^* are equimeasurable on E and $(0, \infty)$, respectively. (The function f^* is called the nonincreasing rearrangement of f.)

28. Let E be a measurable set in $\mathbb{R}^n$ with $|E| < \infty$. Suppose that $f > 0$ a.e. in E and $f, \log f \in L^1(E)$. Prove that

$$\lim_{p \to 0+} \left(\frac{1}{|E|} \int_E f^p \right)^{1/p} = \exp\left(\frac{1}{|E|} \int_E \log f \right).$$

(Start by using Theorem 5.36 to show that $\int_E f^p \to |E|$ as $p \to 0+$. Note that $(f^p - 1)/p \to \log f$.)

29. Let f be measurable, nonnegative, and finite a.e. in a set E. Prove that for any nonnegative constant c,

$$\int_E e^{cf(x)}\,dx = |E| + c\int_0^\infty e^{c\alpha}\omega_f(\alpha)\,d\alpha.$$

Deduce that $e^{cf} \in L(E)$ if $|E| < \infty$ and there exist constants C_1 and c_1 such that $c_1 > c$ and $\omega_f(\alpha) \le C_1 e^{-c_1\alpha}$ for all $\alpha > 0$. We will study such an exponential integrability property in Section 14.5.

6. Repeated Integration

1.

(a) Let E be a measurable subset of $\mathbb{R}^2$ such that for almost every $x \in \mathbb{R}^1$, $\{y : (x, y) \in E\}$ has $\mathbb{R}^1$-measure zero. Show that E has measure zero and that for almost every $y \in \mathbb{R}^1$, $\{x : (x, y) \in E\}$ has measure zero.

(b) Let $f(x, y)$ be nonnegative and measurable in $\mathbb{R}^2$. Suppose that for almost every $x \in \mathbb{R}^1$, $f(x, y)$ is finite for almost every y. Show that for almost every $y \in \mathbb{R}^1$, $f(x, y)$ is finite for almost every x.

2. If f and g are measurable in $\mathbb{R}^n$, show that the function $h(x,y) = f(x)g(y)$ is measurable in $\mathbb{R}^n \times \mathbb{R}^n$. Deduce that if E_1 and E_2 are measurable subsets of $\mathbb{R}^n$, then their Cartesian product $E_1 \times E_2 = \{(x,y) : x \in E_1, y \in E_2\}$ is measurable in $\mathbb{R}^n \times \mathbb{R}^n$, and $|E_1 \times E_2| = |E_1||E_2|$. As usual in measure theory, $0 \cdot \infty$ and $\infty \cdot 0$ are interpreted as 0.

3. Let f be measurable and finite a.e. on $[0,1]$. If $f(x) - f(y)$ is integrable over the square $0 \leq x \leq 1$, $0 \leq y \leq 1$, show that $f \in L[0,1]$.

4. Let f be measurable and periodic with period 1: $f(t+1) = f(t)$. Suppose that there is a finite c such that

$$\int_0^1 |f(a+t) - f(b+t)|\, dt \leq c$$

for all a and b. Show that $f \in L[0,1]$. (Set $a = x$, $b = -x$, integrate with respect to x, and make the change of variables $\xi = x + t$, $\eta = -x + t$.)

5.

(a) If f is nonnegative and measurable on E and $\omega(y) = |\{x \in E : f(x) > y\}|$, $y > 0$, use Tonelli's theorem to prove that $\int_E f = \int_0^\infty \omega(y)\,dy$. (By definition of the integral, $\int_E f = |R(f,E)| = \iint_{R(f,E)} dx\,dy$. Use the observation in the proof of Theorem 6.11 that $\{x \in E : f(x) \geq y\} = \{x : (x,y) \in R(f,E)\}$, and recall that $\omega(y) = |\{x \in E : f(x) \geq y\}|$ unless y is a point of discontinuity of ω.)

(b) Deduce from this special case the general formula

$$\int_E f^p = p\int_0^\infty y^{p-1}\omega(y)\,dy \qquad (f \geq 0,\ 0 < p < \infty).$$

6. For $f \in L(\mathbb{R}^1)$, define the Fourier transform $\hat{f}$ of f by

$$\hat{f}(x) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} f(t) e^{-ixt}\, dt \qquad (x \in \mathbb{R}^1).$$

(For a complex-valued function $F = F_0 + iF_1$ whose real and imaginary parts F_0 and F_1 are integrable, we define $\int F = \int F_0 + i \int F_1$.) Show that if f and g belong to $L(\mathbb{R}^1)$, then

$$\widehat{f * g}(x) = 2\pi \hat{f}(x) \hat{g}(x).$$

7. Let F be a closed subset of $\mathbb{R}^1$ and let $\delta(x) = \delta(x, F)$ be the corresponding distance function. If $\lambda > 0$ and f is nonnegative and integrable over the complement of F, prove that the function

$$\int_{\mathbb{R}^1} \frac{\delta^\lambda(y) f(y)}{|x-y|^{1+\lambda}}\, dy$$

is integrable over F and so is finite a.e. in F. (In case $f = \chi_{(a,b)}$, this reduces to Theorem 6.17.)

8. Under the hypotheses of Theorem 6.17 and assuming that $b - a < 1$, prove that the function

$$M_0(x) = \int_a^b \left[\log \frac{1}{\delta(y)}\right]^{-1} |x - y|^{-1}\, dy$$

is finite a.e. in F.

9.

(a) Show that $M_\lambda(x; F) = +\infty$ if $x \notin F$, $\lambda > 0$.

(b) Let $F = [c, d]$ be a closed subinterval of a bounded open interval $(a, b) \subset \mathbb{R}^1$, and let M_λ be the corresponding Marcinkiewicz integral, $\lambda > 0$. Show that M_λ is finite for every $x \in (c, d)$ and that $M_\lambda(c) = M_\lambda(d) = \infty$. Show also that $\int_F M_\lambda \leq \lambda^{-1}|G|$, where $G = (a, b) - [c, d]$.

10. Let v_n be the volume of the unit ball in $\mathbb{R}^n$. Show by using Fubini's theorem that

$$v_n = 2v_{n-1} \int_0^1 (1-t^2)^{(n-1)/2}\, dt.$$

(We also observe that by setting $w = t^2$, the integral is a multiple of a classical β-function and so can be expressed in terms of the Γ-function: $\Gamma(s) = \int_0^\infty e^{-t} t^{s-1}\, dt$, $s > 0$.)

11. Use Fubini's theorem to prove that

$$\int_{\mathbb{R}^n} e^{-|x|^2}\,dx = \pi^{n/2}.$$

(For $n = 1$, write $(\int_{-\infty}^{+\infty} e^{-x^2}\,dx)^2 = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} e^{-x^2-y^2}\,dx\,dy$ and use polar coordinates. For $n > 1$, use the formula $e^{-|x|^2} = e^{-x_1^2}\cdots e^{-x_n^2}$ and Fubini's theorem to reduce to the case $n = 1$.)

12.

(a) Give an example that shows that the projection onto the x-axis of a measurable subset of the plane may not be linearly measurable.

(b) Show that if E is either an open or closed set in the plane, then its projection onto the x-axis is linearly measurable.

(For (b), the projection of an open set is open, and the projection of a compact set is compact. Any closed set is a countable union of compact sets.)

13. Let $f \in L(-\infty, \infty)$, and let $h > 0$ be fixed. Prove that

$$\int_{-\infty}^{\infty} \left(\frac{1}{2h} \int_{x-h}^{x+h} f(y)\, dy \right) dx = \int_{-\infty}^{\infty} f(x)\, dx.$$

7. Differentiation

1. Let f be measurable in $\mathbb{R}^n$ and different from zero in some set of positive measure. Show that there is a positive constant c such that $f^*(x) \geq c|x|^{-n}$ for $|x| \geq 1$.

2. Let $\phi(x)$, $x \in \mathbb{R}^n$, be a bounded measurable function such that $\phi(x) = 0$ for $|x| \geq 1$ and $\int \phi = 1$. For $\varepsilon > 0$, let $\phi_\varepsilon(x) = \varepsilon^{-n}\phi(x/\varepsilon)$. ($\phi_\varepsilon$ is called an approximation to the identity.) If $f \in L(\mathbb{R}^n)$, show that

$$\lim_{\varepsilon \to 0}(f * \phi_\varepsilon)(x) = f(x)$$

in the Lebesgue set of f. (Note that $\int \phi_\varepsilon = 1$, $\varepsilon > 0$, so that

$$(f * \phi_\varepsilon)(x) - f(x) = \int [f(x-y) - f(x)]\phi_\varepsilon(y)\, dy.$$

Use Theorem 7.16.)

3. Show that the conclusion of Lemma 7.4 remains true for the case of two dimensions if instead of being squares, the sets Q covering E are rectangles with x-dimension equal to h and y-dimension equal to h^2. (Of course, h varies with Q and the rectangles have edges parallel to the coordinate axes.) Show that the conclusion of Theorem 7.2 remains valid for $n = 2$ if the cubes Q are replaced by rectangles of this type that are centered at x and with $h \to 0$.

Show that the same conclusions are valid for rectangles whose x-dimension is h and whose y-dimension is any fixed increasing function of h. Generalize this to higher dimensions.

4. If E_1 and E_2 are measurable subsets of $\mathbb{R}^1$ with $|E_1| > 0$ and $|E_2| > 0$, prove that the set $\{x : x = x_1 - x_2,\ x_1 \in E_1,\ x_2 \in E_2\}$ contains an interval. (cf. Lemma 3.37.)

5. Let f be of bounded variation on $[a, b]$. If $f = g + h$, where g is absolutely continuous and h is singular, show that

$$\int_a^b \phi \, df = \int_a^b \phi f' \, dx + \int_a^b \phi \, dh$$

for any continuous ϕ.

6. Show that if $\alpha > 0$, then x^α is absolutely continuous on every bounded subinterval of $[0, \infty)$.

7. Prove that f is absolutely continuous on $[a, b]$ if and only if given $\varepsilon > 0$, there exists $\delta > 0$ such that $|\sum_i [f(b_i) - f(a_i)]| < \varepsilon$ for every finite collection $\{[a_i, b_i]\}$ of nonoverlapping subintervals of $[a, b]$ with $\sum_i (b_i - a_i) < \delta$.

8. Prove the following converse of Theorem 7.31: If f is of bounded variation on $[a, b]$, and if the function $V(x) = V[a, x]$ is absolutely continuous on $[a, b]$, then f is absolutely continuous on $[a, b]$.

9. If f is of bounded variation on $[a, b]$, show that

$$\int_a^b |f'| \leq V[a, b].$$

Show that if equality holds in this inequality, then f is absolutely continuous on $[a, b]$. (For the second part, use Theorems 2.2(ii) and 7.24 to show that $V(x)$ is absolutely continuous and then use the result of Exercise 8.)

10.

(a) Show that if f is absolutely continuous on $[a, b]$ and Z is a subset of $[a, b]$ of measure zero, then the image set defined by $f(Z) = \{w : w = f(z),\ z \in Z\}$ also has measure zero. Deduce that the image under f of any measurable subset of $[a, b]$ is measurable. (Compare Theorem 3.33.) (Hint: use the fact that the image of an interval $[a_i, b_i]$ is an interval of length at most $V(b_i) - V(a_i)$.)

(b) Give an example of a strictly increasing Lipschitz continuous function f and a set Z with measure 0 such that $f^{-1}(Z)$ does not have measure 0 (and consequently, f^{-1} is not absolutely continuous). (Let $f^{-1}(x) = x + C(x)$ on $[0, 1]$, where $C(x)$ is the Cantor–Lebesgue function.)

11. Prove the following result concerning changes of variable. Let $g(t)$ be monotone increasing and absolutely continuous on $[\alpha, \beta]$ and let f be integrable on $[a, b]$, $a = g(\alpha)$, $b = g(\beta)$. Then $f(g(t))g'(t)$ is measurable and integrable on $[\alpha, \beta]$, and

$$\int_a^b f(x)\,dx = \int_\alpha^\beta f(g(t))g'(t)\,dt.$$

(Consider the cases when f is the characteristic function of an interval, an open set, etc.)

12. Use Jensen's inequality to prove that if $a, b \geq 0$, $p, q > 1$, $(1/p) + (1/q) = 1$, then

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

More generally, show that

$$a_1 \cdots a_N \leq \sum_{j=1}^{N} \frac{a_j^{p_j}}{p_j},$$

where $a_j \geq 0$, $p_j > 1$, $\sum_{j=1}^{N} (1/p_j) = 1$. (Write $a_j = e^{x_j/p_j}$ and use the convexity of e^x.)

13. Prove Theorem 7.36.

14. Prove that ϕ is convex on (a, b) if and only if it is continuous and

$$\phi\left(\frac{x_1 + x_2}{2}\right) \leq \frac{\phi(x_1) + \phi(x_2)}{2}$$

for $x_1, x_2 \in (a, b)$.

15. Theorem 7.43 shows that a convex function is the indefinite integral of a monotone increasing function. Prove the converse: If $\phi(x) = \int_a^x f(t)\,dt + \phi(a)$ in (a, b) and f is monotone increasing, then ϕ is convex in (a, b). (Use Exercise 14.)

16. Show that the formula

$$\int_{-\infty}^{+\infty} fg' = -\int_{-\infty}^{+\infty} f'g$$

for integration by parts may not hold if f is of bounded variation on $(-\infty, +\infty)$ and g is infinitely differentiable with compact support. (Let f be the Cantor–Lebesgue function on $[0, 1]$, and let $f = 0$ elsewhere.)

17. A sequence $\{\phi_k\}$ of set functions is said to be uniformly absolutely continuous if given $\varepsilon > 0$, there exists $\delta > 0$ such that if E satisfies $|E| < \delta$, then $|\phi_k(E)| < \varepsilon$ for all k. If $\{f_k\}$ is a sequence of integrable functions on $(0, 1)$ which converges pointwise a.e. to an integrable f, show that $\int_0^1 |f - f_k| \to 0$ if and only if the indefinite integrals of the f_k are uniformly absolutely continuous. (cf. Exercise 23 of Chapter 10.)

18. Prove the following set-theoretic result related to the simple Vitali covering lemma. If $\mathcal{C} = \{Q\}$ is a collection of cubes all contained in a fixed bounded set in $\mathbb{R}^n$, then there is a countable subcollection $\{Q_k\}$ of disjoint cubes in $\mathcal{C}$ such that every $Q \in \mathcal{C}$ is contained in some Q_k^*, where Q_k^* denotes the cube concentric with Q_k of edge length 5 times that of Q_k.

Deduce the measure-theoretic consequence (cf. Lemma 7.4) that if a set E is covered by such a collection $\mathcal{C}$ of cubes, then there exist $\beta > 0$, depending only on n, and a finite number of disjoint cubes $Q_1, \ldots, Q_N$ in $\mathcal{C}$ such that $\beta |E|_e \leq \sum_{k=1}^{N} |Q_k|$.

Formulate analogues of these facts for a collection of balls in $\mathbb{R}^n$.

19. Use Exercise 18 to prove Lemma 7.9.

20.

(a) Let $f(x)$ be defined for all $x \in \mathbb{R}^n$ by $f(x) = 0$ if every coordinate of x is rational, and $f(x) = 1$ otherwise. Describe the set of all x at which $\frac{1}{|Q|}\int_Q f$ has a limit as $Q \searrow x$ and describe all Lebesgue points of f.

(b) Give an example of a bounded function f on $(-\infty, \infty)$ with the following properties: f is continuous except at a single point x_0; $(d/dx)\int_0^x f = f(x)$ for all x (in particular when $x = x_0$); x_0 is not a Lebesgue point of f.

21. For $x \in \mathbb{R}^n$ and $0 < \alpha < n$, define $f(x) = |x|^{-\alpha}\chi_{\{|x|<1\}}(x)$. Show that its maximal function $f^*(x)$ is bounded both above and below by positive constants (depending only on α and n) times $(|x|^\alpha + |x|^n)^{-1}$.

22. In order to better understand Theorem 7.24 and its proof on an intuitive level, it may be helpful to sketch the graphs of a particularly simple continuously differentiable function $f(x)$ and its variation $V(x)$ on the same set of axes, and then to compare f' and V'. Draw both graphs for $f(x) = \sin x$ on $[0, 2\pi]$ or more generally for any f defined on a closed interval of finite length whose graph is a finite union of smooth monotone arcs.

23. Show that a convex function on (a, b) cannot attain a maximum on (a, b) unless it is constant. Is the same true for a local maximum?

24. Suppose that ϕ is continuous on (a, b), ϕ' exists and is finite on (a, b) except at a single point ξ, and ϕ' is increasing on $(a, \xi) \cup (\xi, b)$. Show that ϕ is convex on (a, b).

25. Let $1 \leq p < \infty$. Show that $x^p \log(1+x)$ and $x^p(1+\log^+ x)$ are convex on $(0,\infty)$. (For the second function, Exercise 24 may be helpful.) See also Exercise 28 in Chapter 8.

26. Show that the function defined in (7.48) is Lipschitz continuous on $\mathbb{R}^2$.

27. Verify the uniqueness of A in (7.49) in the sense described immediately after (7.49). (Show that if 0 is a point of density of a measurable set $E \subset \mathbb{R}^n$ and B is a nonzero vector in $\mathbb{R}^n$, then there is a sequence $\{h_j\} \subset E$ such that $|h_j| \to 0$, $|h_j| \neq 0$, and the angle between h_j and B is bounded away from $\pi/2$ uniformly in j.)

28. Let $f \in \mathrm{Lip}(\mathbb{R}^n)$ and T be a nonsingular linear transformation of $\mathbb{R}^n$. Show that the function $f \circ T$ defined by $(f \circ T)(x) = f(Tx)$ belongs to $\mathrm{Lip}(\mathbb{R}^n)$ and that for a.e. $x \in \mathbb{R}^n$, $\nabla(f \circ T)(x) = T^*[(\nabla f)(Tx)]$, where T^* denotes the adjoint (transpose) of T.

29.

(a) In one dimension, give an example of a Lipschitz function f that satisfies the smoothness condition $f(x_0+h)+f(x_0-h)-2f(x_0)=o(|h|)$ as $h\to 0$ at a point x_0 but for which $f'(x_0)$ does not exist.

(b) Show that any (finite) f that satisfies the smoothness condition in part (a) at a point x_0 where f has a local extremum is differentiable at x_0 and $f'(x_0)=0$. (Express the ratio $[f(x_0+h)+f(x_0-h)-2f(x_0)]/h$ in terms of right- and left-hand difference quotients of f at x_0.)

(c) Suppose that a function f satisfies the smoothness condition in part (a) for every x in an interval (a,b) of the real line and that f is also continuous in (a,b). Show that f' exists and is finite in a dense subset of (a,b). (If $[a',b']\subset(a,b)$, apply the result in part (b) to the function $f(x)-L(x)$ where L is the chord equal to f at $x=a',b'$.)

(d) Prove the following generalization of the classical mean value theorem: if f is continuous in an interval $[a,b]\subset(-\infty,\infty)$ and smooth (in the sense of part (a)) in (a,b), then there exists $c\in(a,b)$ such that $f(b)-f(a)=f'(c)(b-a)$.

30. Let H be a measurable set in $\mathbb{R}^n$ that has 0 as a point of density. For $\rho > 0$, let H_ρ be the intersection of H with the ball B_ρ of radius ρ centered at 0.

(a) Show that the set $\{2x - y : x, y \in H_\rho\}$ covers B_ρ if ρ is sufficiently small. Deduce that the set $\{x + h : x, x - h \in H_\rho,\ h \in \mathbb{R}^n\}$ covers B_ρ if ρ is small. (For the first part, compare Lemma 3.37. Recall from Theorem 3.35 that if E is a measurable set in $\mathbb{R}^n$, then the set $2E$ defined by $2E = \{2x : x \in E\}$ has measure $|2E| = 2^n|E|$. Note that $2H_\rho$ and the set obtained by translating H_ρ by any fixed point in B_ρ are both subsets of $B_{2\rho}$.)

(b) While the result in part (a) is adequate for the purpose of proving Theorem 7.63, it can be improved and generalized. If r, s are real numbers with $0 < |s| \leq |r|$, show that the set $\{rx + sy : x, y \in H_\rho\}$ covers $B_{|r|\rho}$ if ρ is sufficiently small. (Argue as for part (a), but consider only translations of an appropriate subset of $-sH_\rho$.)

31. Let f be a finite measurable function on $\mathbb{R}^n$ that satisfies $f(x+h)+f(x-h)-2f(x)=O(1)$ uniformly in x, h for $|h|<\delta$, that is, there are positive constants A, δ such that $|f(x+h)+f(x-h)-2f(x)|\leq A$ if $x, h \in \mathbb{R}^n$ and $|h|<\delta$. Show that f is bounded on every bounded set in $\mathbb{R}^n$. (First show that f is bounded on some ball in $\mathbb{R}^n$. To do so, note that there is a measurable set E with $|E|>0$ on which f is bounded. Pick a point of density of E and apply Exercise 30.) Generalize the result to functions f that satisfy $f(x+ay)+f(x+by)-2f(x)=O(1)$ for fixed real $a, b \neq 0$.

32. Complete the proof of the extension Theorem 7.64 by verifying the properties listed in the last line of its proof for the function f_1 defined in (7.65).

33. Let f be defined and Lipschitz continuous on a measurable set $E \subset \mathbb{R}^n$. Show that f has an approximate differential relative to E at almost every point of E.

8. L^p Classes

1. For complex-valued measurable f, $f = f_1 + if_2$ with f_1 and f_2 real-valued and measurable, we have $\int_E f = \int_E f_1 + i\int_E f_2$ by definition. Prove that $\int_E f$ is finite if and only if $\int_E |f|$ is finite, and $\left|\int_E f\right| \le \int_E |f|$. (Note that $\left|\int_E f\right| = [(\int_E f_1)^2 + (\int_E f_2)^2]^{1/2}$, and use the fact that $(a^2+b^2)^{1/2} = a\cos\alpha + b\sin\alpha$ for an appropriate α, while $(a^2+b^2)^{1/2} \ge |a\cos\alpha + b\sin\alpha|$ for all α.)

2. Prove the converse of Hölder's inequality for $p = 1$ and ∞. Show also that for $1 \leq p \leq \infty$, a real-valued measurable f belongs to $L^p(E)$ if $fg \in L^1(E)$ for every $g \in L^{p'}(E)$, $1/p+1/p' = 1$. The negation is also of interest: if $f \notin L^p(E)$, then there exists $g \in L^{p'}(E)$ such that $fg \notin L^1(E)$. (To verify the negation, construct g of the form $\sum a_k g_k$ for appropriate a_k and g_k, with g_k satisfying $\int_E f g_k \to +\infty$.)

3. Prove Theorems 8.12 and 8.13. Show that Minkowski's inequality for series fails when $p < 1$.

4. Let f and g be real-valued and not identically 0 (i.e., neither function equals 0 a.e.), and let $1 < p < \infty$. Prove that equality holds in the inequality $\left|\int fg\right| \leq \|f\|_p \|g\|_{p'}$ if and only if fg has constant sign a.e. and $|f|^p$ is a multiple of $|g|^{p'}$ a.e.

If $\|f + g\|_p = \|f\|_p + \|g\|_p$ and $g \neq 0$ in Minkowski's inequality, show that f is a multiple of g.

Find analogues of these results for the spaces l^p.

5. For $0 < p \leq \infty$ and $0 < |E| < +\infty$, define

$$N_p[f] = \left(\frac{1}{|E|} \int_E |f|^p \right)^{1/p},$$

where $N_\infty[f]$ means $\|f\|_\infty$. Prove that if $p_1 < p_2$, then $N_{p_1}[f] \leq N_{p_2}[f]$.

Prove also that if $1 \leq p \leq \infty$, then $N_p[f+g] \leq N_p[f] + N_p[g]$, $(1/|E|) \int_E |fg| \leq N_p[f] N_{p'}[g]$, $1/p + 1/p' = 1$, and that $\lim_{p \to \infty} N_p[f] = \|f\|_\infty$. Thus, N_p behaves like $\|\cdot\|_p$ but has the advantage of being monotone in p. Recall Exercise 28 of Chapter 5.

6.

(a) Let $1 \leq p_i, r \leq \infty$ and $\sum_{i=1}^{k} \frac{1}{p_i} = \frac{1}{r}$. Prove the following generalization of Hölder's inequality:

$$\|f_1 \cdots f_k\|_r \leq \|f_1\|_{p_1} \cdots \|f_k\|_{p_k}.$$

(See also Exercise 12 of Chapter 7.)

(b) Let $1 \leq p < r < q \leq \infty$ and define $\theta \in (0,1)$ by $\frac{1}{r} = \frac{\theta}{p} + \frac{1-\theta}{q}$. Prove the interpolation estimate

$$\|f\|_r \leq \|f\|_p^{\theta} \|f\|_q^{1-\theta}.$$

In particular, if $A = \max\{\|f\|_p, \|f\|_q\}$, then $\|f\|_r \leq A$.

7. Show that when $0 < p < 1$, the neighborhoods $\{f : \|f\|_p < \varepsilon\}$ of zero in $L^p(0,1)$ are not convex. (Let $f = \chi_{(0,\varepsilon^p)}$, and $g = \chi_{(\varepsilon^p, 2\varepsilon^p)}$. Show that $\|f\|_p = \|g\|_p = \varepsilon$, but that $\|\frac{1}{2}f + \frac{1}{2}g\|_p > \varepsilon$.)

8. Prove the following integral version of Minkowski's inequality for $1 \leq p < \infty$ and a measurable function $f(x, y)$:

$$\left(\int \left(\int |f(x,y)|\, dx \right)^p dy \right)^{1/p} \leq \int \left(\int |f(x,y)|^p dy \right)^{1/p} dx.$$

(For $1 < p < \infty$, note that the pth power of the left-hand side equals

$$\iint \left(\int |f(z,y)|\, dz \right)^{p-1} |f(x,y)|\, dx\, dy.$$

Integrate first with respect to y and apply Hölder's inequality.)

9. If f is real-valued and measurable on E, $|E| > 0$, define its essential infimum on E by

$$\operatorname*{ess\,inf}_E f = \sup\{\alpha : |\{x \in E : f(x) < \alpha\}| = 0\}.$$

If $f \geq 0$, show that $\operatorname{ess\,inf}_E f = (\operatorname{ess\,sup}_E 1/f)^{-1}$.

10. Prove that $L^\infty(E)$ is not separable for any E with $|E| > 0$. (Construct a sequence of decreasing subsets of E whose measures strictly decrease. Consider the characteristic functions of the class of sets obtained by taking all possible unions of the differences of these subsets.)

11. If $f_k \to f$ in L^p, $1 \le p < \infty$, $g_k \to g$ pointwise, and $\|g_k\|_\infty \le M$ for all k, prove that $f_k g_k \to fg$ in L^p.

12. Let $f, \{f_k\} \in L^p$, $0 < p \leq \infty$. Show that if $\|f - f_k\|_p \to 0$, then $\|f_k\|_p \to \|f\|_p$. Conversely, if $f_k \to f$ a.e. and $\|f_k\|_p \to \|f\|_p$, $0 < p < \infty$, show that $\|f - f_k\|_p \to 0$. Show that the converse may fail for $p = \infty$. (For the converse when $0 < p < \infty$, note that $|f - f_k|^p \leq c(|f|^p + |f_k|^p)$ with $c = \max\{2^{p-1}, 1\}$; then apply, for example, the sequential version of Lebesgue's dominated convergence theorem given in Exercise 23 of Chapter 5.)

13. Suppose that $f_k \to f$ a.e. and that $f_k, f \in L^p$, $1 < p \leq \infty$. If $\|f_k\|_p \leq M < +\infty$, show that $\int f_k g \to \int fg$ for all $g \in L^{p'}$, $1/p + 1/p' = 1$. Show that the result is false if $p = 1$. (When $p > 1$, use Egorov's theorem in case the domain of integration has finite measure.)

14. Verify that the following systems are orthogonal:

(a) $\{\frac{1}{2}, \cos x, \sin x, \ldots, \cos kx, \sin kx, \ldots\}$ on any interval of length 2π.

(b) $\{e^{2\pi ikx/(b-a)};\ k = 0, \pm 1, \pm 2, \ldots\}$ on (a, b).

15. If $f \in L^2(0, 2\pi)$, show that

$$\lim_{k\to\infty} \int_0^{2\pi} f(x) \cos kx \, dx = \lim_{k\to\infty} \int_0^{2\pi} f(x) \sin kx \, dx = 0.$$

Prove that the same is true if $f \in L^1(0, 2\pi)$. (This last statement is the Riemann–Lebesgue lemma. To prove it, approximate f in L^1 norm by L^2 functions. See Theorem 12.21.)

16. A sequence $\{f_k\}$ in L^p is said to converge weakly in L^p to a function f (belonging to L^p) if $\int f_k g \to \int fg$ for all $g \in L^{p'}$. Prove that if $f_k \to f$ in L^p norm, $1 \leq p \leq \infty$, then $\{f_k\}$ converges weakly in L^p to f. Note by Exercise 15 that the converse is not true. See Exercise 28 of Chapter 10.

17. Suppose that $f_k, f \in L^2$ and that $\int f_k g \to \int fg$ for all $g \in L^2$ (i.e., $\{f_k\}$ converges weakly in L^2 to f). If $\|f_k\|_2 \to \|f\|_2$, show that $f_k \to f$ in L^2 norm. The same is true for L^p, $1 < p < \infty$, by a 1913 result of Radon.

18. Prove the parallelogram law for L^2:

$$\|f+g\|_2^2 + \|f-g\|_2^2 = 2\|f\|_2^2 + 2\|g\|_2^2.$$

Is this true for L^p when $p \neq 2$? The geometric interpretation is that the sum of the squares of the lengths of the diagonals of a parallelogram equals the sum of the squares of the edge lengths.

19. Prove that a finite dimensional Hilbert space is isometric with $\mathbb{R}^n$ for some n.

20. Construct a function in $L^1(-\infty, +\infty)$ that is not in $L^2(a, b)$ for any $a < b$. (Let $g(x) = x^{-1/2}$ on $(0, 1)$ and $g(x) = 0$ elsewhere, so that $\int_{-\infty}^{+\infty} g = 2$. Consider the function $f(x) = \sum a_k g(x - r_k)$, where $\{r_k\}$ is the rational numbers and $\{a_k\}$ satisfies $a_k > 0$, $\sum a_k < +\infty$.)

21. If $f \in L^p(\mathbb{R}^n)$, $0 < p < \infty$, show that

$$\lim_{Q \searrow x} \frac{1}{|Q|} \int_Q |f(y) - f(x)|^p dy = 0 \quad \text{a.e.}$$

Note by Exercise 5 that if this condition holds for a given p, then it also holds for all smaller p.

22. Let $\{\phi_k\}$ be a complete orthonormal system in L^2 and let $m = \{m_k\}$ be a fixed bounded sequence of numbers. If $f \in L^2$, $f \sim \sum c_k \phi_k$, define Tf by $Tf \sim \sum m_k c_k \phi_k$. Such an operator is called a Fourier multiplier operator. Show that T is bounded on L^2, that is, that there is a constant c independent of f such that $\|Tf\|_2 \leq c\|f\|_2$ for all $f \in L^2$. Show also that the smallest possible choice for c is $\|m\|_{l^\infty}$.

23. Show that every subset Λ of a separable metric space (M, d) is separable. (Let $D = \{f_k\}$ be a countable dense set in M, and for $j = 1, 2, \ldots$, define $D_j = \{f \in D : \inf_{\lambda \in \Lambda} d(\lambda, f) < 1/j\}$. If $f_k \in D_j$, pick $\lambda_{k,j} \in \Lambda$ with $d(\lambda_{k,j}, f_k) < 1/j$ and show that $\{\lambda_{k,j}\}$ is dense in Λ.)

24. Let E be a measurable set in $\mathbb{R}^n$ with $0 < |E| < \infty$. Construct an orthogonal system $\{\phi_j\}_{j=0}^{\infty}$ in $L^2(E)$ with $\phi_0 = 1$ everywhere in E. (Use Exercise 32 of Chapter 3 with $\theta = 1/2$, and choose ϕ_j for $j \geq 1$ to be appropriate simple functions with values ± 1.)

25. If f is a measurable function on $\mathbb{R}^n$, define $\langle f\rangle = \sup_{\alpha>0}\alpha|\{|f| > \alpha\}|$, and recall that f belongs to weak $L^1(\mathbb{R}^n)$ if and only if $\langle f\rangle < \infty$. Show that weak $L^1(\mathbb{R}^n)$ has all the properties of a Banach space with respect to $\langle\cdot\rangle$ except the triangle inequality. Show however that there is a constant $\kappa > 1$ such that the quasi-triangle inequality $\langle f+g\rangle \le \kappa(\langle f\rangle + \langle g\rangle)$ holds for all measurable f, g. (To show that κ cannot be 1, consider the case of one dimension and the functions $f = \chi_{[0,1/2]} + 2\chi_{(1/2,1]}$, $g = 2\chi_{[0,1/2]} + \chi_{(1/2,1]}$.)

26. Show that $\liminf_{p\to\infty} \|f\|_{L^p(E)} \geq \|f\|_{L^\infty(E)}$ even if $|E| = \infty$.

27.

(a) Prove Minkowski's inequality for infinite series:

$$\left\| \sum_{k=1}^{\infty} |f_k| \right\|_p \leq \sum_{k=1}^{\infty} \|f_k\|_p, \qquad 1 \leq p \leq \infty.$$

(b) Show that in part (a), the opposite inequality holds if $0 < p \leq 1$:

$$\sum_{k=1}^{\infty} \|f_k\|_p \leq \left\| \sum_{k=1}^{\infty} |f_k| \right\|_p, \qquad 0 < p \leq 1.$$

(For (b), assuming that $\sum |f_k|$ is positive and finite a.e., multiply and divide each $|f_k|^p$ in the summation on the left side of the inequality by $(\sum |f_k|)^{p(1-p)}$ and apply Hölder's inequality with exponents $q = 1/p$ and $q' = 1/(1-p)$.)

28. Let $\phi(t)$ be a continuous function on $[0,\infty)$ that is positive, increasing, and convex on $(0,\infty)$ and that satisfies $\phi(0)=0$, $\lim_{t\to\infty}\phi(t)=\infty$, and $|\phi(2t)|\le c|\phi(t)|$ for some constant c independent of t. For example, the function $\phi(t)=t(1+\log^+ t)$ has these properties (see Exercise 25 of Chapter 7). If E is a measurable set in $\mathbb{R}^n$, define the Orlicz space $L^\phi(E)$ to be the collection of all measurable f on E that are finite a.e. in E and satisfy $\phi(|f|)\in L(E)$. Show that $L^\phi(E)$ is a Banach space with norm

$$\|f\|_{L^\phi(E)}=\inf\left\{\lambda>0:\int_E\phi\left(\frac{|f|}{\lambda}\right)dx\le 1\right\}.$$

In case $\phi(t)=t(1+\log^+ t)$, the class is often denoted $L\log L(E)$ (see Exercise 22 of Chapter 9 for a result about functions in $L\log L(\mathbb{R}^n)$).

29. Let $1 \leq p < \infty$ and $f \in L^p(\mathbb{R}^n)$. Show that $\|f(x+h) - f(x)\|_p$ (where the norm is taken with respect to x) is a uniformly continuous function of h. Is the same true when $0 < p < 1$?

30. Let $1 \le p < \infty$ and E be a measurable set in $\mathbb{R}^n$.

(a) Prove that if f_1, f_2, g_1, g_2 are nonnegative and measurable on E, then

$$\left(\int_E [(f_1+f_2)^p + (g_1+g_2)^p]\,dx\right)^{1/p} \le \left(\int_E (f_1^p + g_1^p)\,dx\right)^{1/p} + \left(\int_E (f_2^p + g_2^p)\,dx\right)^{1/p}.$$

(b) If the right side in part (a) is replaced by

$$\left(\int_E (f_1^p + f_2^p)\,dx\right)^{1/p} + \left(\int_E (g_1^p + g_2^p)\,dx\right)^{1/p},$$

is the resulting inequality true?

(c) If $\{f_{i,j}\}_{i,j=1}^{\infty}$ are measurable functions on E, show that

$$\left[\int_E \sum_j \left(\sum_i |f_{i,j}(x)|\right)^p dx\right]^{1/p} \le \sum_i \left[\int_E \left(\sum_j |f_{i,j}(x)|^p\right) dx\right]^{1/p}.$$

(For $i = 1, \ldots, N$ (N finite), consider the sequences $F_i = \{|f_{i,j}|\}_j$ and note that $\|\sum_{i=1}^N F_i\|_{l^p} = (\sum_j (\sum_{i=1}^N |f_{i,j}|)^p)^{1/p}$.)

31. Let $a = \{a_k\}$ be a sequence of real or complex numbers. Show that $\|a\|_p \leq \|a\|_1$ if $1 \leq p \leq \infty$ and more generally that $\|a\|_p \leq \|a\|_q$ if $0 < q \leq p \leq \infty$. (If $1 \leq p < \infty$, the inequality $|a_1|^p + |a_2|^p \leq (|a_1| + |a_2|)^p$ may be used together with an induction argument.)

32. For nonnegative measurable functions f and g on $(0,\infty)$, let

$$F(x) = \int_0^\infty f\left(\frac{x}{y}\right) g(y)\,\frac{dy}{y}, \qquad x \in (0,\infty).$$

Also, if $1 \le p < \infty$, set

$$[f]_p = \left(\int_0^\infty f(x)^p\,\frac{dx}{x}\right)^{1/p},$$

and if $p = \infty$, define $[f]_\infty = \operatorname{ess\,sup}_{(0,\infty)} f$. Prove that for $1 \le p \le \infty$, $[F]_p \le [f]_1[g]_p$.

9. Approximations of the Identity and Maximal Functions

1. Use Minkowski's integral inequality (see Exercise 8 of Chapter 8) to prove Theorem 9.1 for $1 < p < \infty$.

2.

(a) Prove Young's Theorem 9.2. (For $f, g \geq 0$ and $p, q, r < \infty$, write

$$(f * g)(x) = \int f(t)^{p/r} g(x-t)^{q/r} \cdot f(t)^{p(1/p-1/r)} \cdot g(x-t)^{q(1/q-1/r)}\, dt,$$

and apply Hölder's inequality for three functions (Exercise 6 of Chapter 8) with exponents r, p_1, and p_2, where $1/p_1 = 1/p - 1/r$, $1/p_2 = 1/q - 1/r$.)

(b) Suppose that the conclusion of Theorem 9.2 holds for three indices $p, q, r > 0$. Show that $1/r = 1/p + 1/q - 1$. (Apply the inequality in the conclusion to the dilated functions $f(\lambda x)$ and $g(\lambda x)$, $\lambda > 0$, and let λ vary. A similar method is used in Exercise 13 of Chapter 14.)

3.

(a) Show that if $f \in L^p(\mathbb{R}^n)$ and $K \in L^{p'}(\mathbb{R}^n)$, $1 \leq p \leq \infty$, $1/p + 1/p' = 1$, then $f * K$ is bounded and continuous in $\mathbb{R}^n$.

(b) Sketch the (trapezoidal) graph of $\chi_I * \chi_J$ where I and J are one-dimensional intervals. Consider also the case when the two intervals are the same.

4.

(a) Show that the function h defined by $h(x) = e^{-1/x^2}$ for $x > 0$ and $h(x) = 0$ for $x \leq 0$ is in C^∞.

(b) Show that the function $g(x) = h(x-a)h(b-x)$, $a < b$, is C^∞ with support $[a, b]$.

(c) Construct a function in $C_0^\infty(\mathbb{R}^n)$ whose support is a ball or an interval.

5. Let G and G_1 be bounded open subsets of $\mathbb{R}^n$ such that $\overline{G_1} \subset G$. Construct a function $h \in C_0^\infty$ such that $h = 1$ in $\overline{G_1}$ and $h = 0$ outside G. (Choose an open G_2 such that $\overline{G_1} \subset G_2$, $\overline{G_2} \subset G$. Let $h = \chi_{G_2} * K$ for a $K \in C^\infty$ with suitably small support and $\int K = 1$.)

6. Prove Theorem 9.4.

7. Let $f \in L^p(-\infty, +\infty)$, $1 \leq p \leq \infty$. Show that the Poisson integral of f, $f(x, y)$, is harmonic in the upper half-plane $y > 0$. (Show that $((\partial^2/\partial x^2) + (\partial^2/\partial y^2))f(x, y) = \int_{-\infty}^{+\infty} f(t)((\partial^2/\partial x^2) + (\partial^2/\partial y^2))P_y(x - t)\, dt$.)

8. (Schur's lemma) For $s, t \geq 0$, let $K(s,t)$ satisfy $K \geq 0$ and $K(\lambda s, \lambda t) = \lambda^{-1} K(s,t)$ for all $\lambda > 0$, and suppose that $\int_0^\infty t^{-1/p} K(1,t)\, dt = \gamma < +\infty$ for some p, $1 \leq p \leq \infty$. For example, $K(s,t) = 1/(s+t)$ has these properties. Show that if

$$(Tf)(s) = \int_0^\infty f(t) K(s,t)\, dt \qquad (f \geq 0),$$

then $\|Tf\|_p \leq \gamma \|f\|_p$. (Note that $K(s,t) = s^{-1} K(1, t/s)$, and therefore $(Tf)(s) = \int_0^\infty f(ts) K(1,t)\, dt$. Now apply Minkowski's integral inequality [see Exercise 8 of Chapter 8].)

9.

(a) The maximal function is defined as $f^*(x) = \sup |Q|^{-1} \int_Q |f|$, where the supremum is taken over cubes Q with center x. Let $f^{**}(x)$ be defined similarly, but with the supremum taken over all cubes Q containing x. Thus, $f^*(x) \leq f^{**}(x)$. Show that there is a positive constant c depending only on the dimension such that $f^{**}(x) \leq cf^*(x)$.

(b) If f^{**} were instead defined to be $\sup |B|^{-1} \int_B |f|$ where the supremum is taken over all balls B containing x, show that there are positive constants c_1 and c_2 depending only on the dimension so that $c_1 f^*(x) \leq f^{**}(x) \leq c_2 f^*(x)$ for all x.

10. Let $T : f \to Tf$ be a function transformation that is sublinear; that is, T has the property that if Tf_1 and Tf_2 are defined, then so is $T(f_1 + f_2)$, and

$$|T(f_1 + f_2)(x)| \leq |(Tf_1)(x)| + |(Tf_2)(x)|.$$

Suppose also that there are constants c_1 and c_2 such that T satisfies $\|Tf\|_\infty \leq c_1\|f\|_\infty$ and $|\{x : |(Tf)(x)| > \alpha\}| \leq c_2\alpha^{-1}\|f\|_1$, $\alpha > 0$. Show that for $1 < p < \infty$, there is a constant c_3 such that $\|Tf\|_p \leq c_3\|f\|_p$. This is a special case of an interpolation result due to Marcinkiewicz. (An example of such a T is the maximal function operator $Tf = f^*$, and the proof in the general case is like that for f^*.)

11. Generalize Theorem 9.6 as follows: Let $f_\varepsilon = f * K_\varepsilon$, $K \in L^1(\mathbb{R}^n)$ and $\int_{\mathbb{R}^n} K = \gamma$. If $f \in L^p(\mathbb{R}^n)$, $1 \leq p < \infty$, show that $\|f_\varepsilon - \gamma f\|_p \to 0$. Derive analogous results for Theorems 9.8, 9.9, and 9.13. (The case $\gamma \neq 0$ follows from the case $\gamma = 1$ by considering $K(x)/\gamma$.)

12. Show that the conclusions of Theorems 9.9 and 9.13 remain true if the assumption that $f \in L^1$ is replaced by $f \in L^p$, $p > 1$.

13. Let $f \in L^p(0,1)$, $1 \le p < \infty$, and for each $k = 1, 2, \ldots$, define a function f_k on $(0,1)$ by letting $I_{k,j} = \{x : (j-1)2^{-k} \le x < j2^{-k}\}$, $j = 1, \ldots, 2^k$, and setting $f_k(x)$ equal to $|I_{k,j}|^{-1} \int_{I_{k,j}} f$ for $x \in I_{k,j}$. Prove that $f_k \to f$ in $L^p(0,1)$ norm. (Exercise 17 of Chapter 7 may be helpful for the case $p = 1$.)

14. Show that Theorem 9.6 and Corollary 9.7 fail for $p = \infty$.

15. Regarding the Dirichlet problem for the upper half-space, more can be said about the behavior of the Poisson integral $f(x, y)$ of $f(x)$ near a point of continuity of $f(x)$. Prove that if $f \in L^p(-\infty, \infty)$, $1 \leq p \leq \infty$, and f is continuous at a point x_0, then $f(x, y) \to f(x_0)$ as (x, y) approaches x_0 unrestrictedly, that is, as $x \to x_0$ and $y \to 0$, $y > 0$.

16. Let $1 \leq p \leq \infty$, $f \in L^p(-\infty, \infty)$, and x_0 be a Lebesgue point of f. Show that the Poisson integral $f(x, y)$ of f converges nontangentially to $f(x_0)$, that is, show that for any $\gamma > 0$, $f(x, y) \to f(x_0)$ as $x \to x_0$ and $y \to 0$ with $|x - x_0| \leq \gamma y$. (Note that the Poisson kernel satisfies $P_y(t + z) \leq C_\gamma P_y(t)$ if $|z| \leq \gamma y$, with C_γ independent of y, t, z.) (See also Theorems 12.42 and 12.64.)

17. Prove that the conclusion of Lemma 9.14 holds without assuming ϕ is continuous provided $\int_a^b \phi(\rho)\, dF(\rho)$ exists. Show, for example, that the conclusion holds if ϕ is monotone and finite on $[a, b]$. (For the second part, recall from Theorem 2.21 that $\int_a^b \phi\, dF$ exists if $\int_a^b F\, d\phi$ does.)

18. Derive the first part of Theorem 9.17 by using the formula in the conclusion of Lemma 9.14 (even though the function ϕ in Theorem 9.17 is not assumed to be continuous). (Use Exercise 17. Note that if $K(x) = \phi(|x|)$ satisfies the hypothesis of Theorem 9.17, then $\phi(|x|) = o(|x|^{-n})$ as $|x| \to 0$ and as $|x| \to \infty$.)

19. Let $K(x)$ be a nonnegative, decreasing, integrable radial function on $\mathbb{R}^n$ (i.e., K satisfies the hypothesis of Theorem 9.17), and let $\int K = 1$. Use Theorem 9.17 to show that if $1 \leq p \leq \infty$ and $f \in L^p(\mathbb{R}^n)$, then $f * K_\varepsilon \to f$ a.e. as $\varepsilon \to 0$. (In case $1 \leq p < \infty$, a proof reminiscent of the proof of Lebesgue's differentiation theorem can be constructed by applying Corollary 9.7.)

20. Show that the conclusion $f * K_\varepsilon \to f$ in Exercise 19 is valid at every Lebesgue point of f. (A proof based on integrating the formula in Lemma 9.14 by parts is possible; cf. Exercise 18.)

21. Let f and g be locally integrable functions on $\mathbb{R}^n$ and suppose that $|f| * |g|$ is finite a.e. in $\mathbb{R}^n$. Prove that $f * g$ is measurable on $\mathbb{R}^n$. (See the argument in the last part of the proof of Theorem 9.1 involving the truncated functions f_N and also truncate g.)

22. As we know from Chapter 7, the maximal function f^* of an $f \in L^1(\mathbb{R}^n)$ may not be locally integrable. Show that if f satisfies the stronger condition $\int_{\mathbb{R}^n} |f|(1 + \log^+ |f|)\, dx < \infty$, then $f^* \in L^1(E)$ for every measurable set E with $|E| < \infty$, and

$$\int_E f^*\, dx \le C \left(|E| + \int_{\mathbb{R}^n} |f| \log^+ |f|\, dx \right),$$

with C independent of f and E. (Let $\omega(\alpha)$ denote the distribution function of f^* relative to E. Write $\int_0^\infty \omega(\alpha)\, d\alpha = \int_0^\gamma + \int_\gamma^\infty$ for $\gamma > 0$. The first integral on the right is bounded by $\gamma|E|$. In the second integral, use the estimate $\omega(\alpha) \le C\alpha^{-1} \int_{\{|f|>\alpha/2\}} |f|$; cf. the proof of Theorem 9.16.)

23.

(a) Show that there is no identity element for the convolution operation on $L^1(\mathbb{R}^n)$, that is, there is no function $k \in L^1(\mathbb{R}^n)$ such that $f * k = f$ a.e. for every $f \in L^1(\mathbb{R}^n)$.

(b) Show that if $f, g \in L^2(\mathbb{R}^n)$, then $f * g$ belongs to $L^\infty(\mathbb{R}^n)$ but not necessarily to $L^2(\mathbb{R}^n)$. See also Lemma 13.49.

10. Abstract Integration

1. Prove Theorem 10.13.

2. A measure space $(\mathcal{S}, \Sigma, \mu)$ is said to be complete if Σ contains all subsets of sets with measure zero; that is, $(\mathcal{S}, \Sigma, \mu)$ is complete if $Y \in \Sigma$ whenever $Y \subset Z$, $Z \in \Sigma$, and $\mu(Z) = 0$. In this case, show that if f is measurable and $g = f$ a.e. (μ), then g is also measurable (cf. Theorem 4.5 and Chapter 3, Exercise 34). Is this true if $(\mathcal{S}, \Sigma, \mu)$ is not complete?

Give an example of an incomplete measure space with a measure that is neither identically infinite nor identically zero.

3. Prove Egorov's Theorem 10.14.

4. If $(\mathcal{S}, \Sigma, \mu)$ is a measure space, and if f and $\{f_k\}$ are measurable and finite a.e. (μ) in a measurable set E, then $\{f_k\}$ is said to converge in μ-measure on E to limit f if

$$\lim_{k\to\infty} \mu\{x \in E : |f(x) - f_k(x)| > \varepsilon\} = 0 \quad \text{for all } \varepsilon > 0.$$

Formulate and prove analogues of Theorems 4.21 through 4.23.

5. Complete the proof of Lemma 10.18.

6.

(a) If $f_1, f_2 \in L(d\mu)$ and $\int_E f_1\, d\mu = \int_E f_2\, d\mu$ for all measurable E, show that $f_1 = f_2$ a.e. (μ).

(b) Prove the uniqueness of f and σ in Theorem 10.40.

(c) Let μ be σ-finite, and let $f_1, f_2 \in L^{p'}(d\mu)$, $1/p + 1/p' = 1$, $1 \leq p \leq \infty$. If $\int f_1 g\, d\mu = \int f_2 g\, d\mu$ for all $g \in L^p(d\mu)$, show that $f_1 = f_2$ a.e. (μ).

7. Prove the integral convergence results in Theorems 10.27 through 10.29 and 10.31.

8. Show that for $1 \leq p < \infty$, the class of simple functions vanishing outside sets of finite measure is dense in $L^p(d\mu)$. See also Exercise 27.

9. The symmetric difference of two sets E_1 and E_2 is defined as

$$E_1 \,\Delta\, E_2 = (E_1 - E_2) \cup (E_2 - E_1).$$

Let $(\mathcal{S}, \Sigma, \mu)$ be a measure space, and identify measurable sets E_1 and E_2 if $\mu(E_1 \Delta E_2) = 0$. Show that Σ is a metric space with distance $d(E_1, E_2) = \mu(E_1 \Delta E_2)$ and that if μ is finite, then $L^p(\mathcal{S}, \Sigma, \mu)$ is separable if and only if Σ is, $1 \leq p < \infty$. (For the sufficiency in the second part, Exercise 8 may be helpful; for the necessity, let $\{f_k\}$ be a countable dense set in $L^p(\mathcal{S}, \Sigma, \mu)$ and consider the sets $\{1/2 < f_k \leq 3/2\}$.)

10. If ϕ is a set function whose Jordan decomposition is $\phi = \overline{V} - \underline{V}$, define

$$\int_E f\, d\phi = \int_E f\, d\overline{V} - \int_E f\, d\underline{V},$$

provided not both integrals on the right are infinite with the same sign. If V is the total variation of ϕ on E, and if $|f| \leq M$, prove that $|\int_E f\, d\phi| \leq MV$.

11. Prove parts (ii)–(iv) of Theorem 10.33.

12. Give an example of a pair of measures ν and μ such that ν is absolutely continuous with respect to μ, but given $\varepsilon > 0$, there is no $\delta > 0$ such that $\nu(A) < \varepsilon$ for every A with $\mu(A) < \delta$. (Thus, the analogue for measures of Theorem 10.34 may fail.)

Prove the analogue of Theorem 10.35 for mutually singular measures ν and μ.

13. Show that the set P of the Hahn decomposition is unique up to null sets. (By a null set for ϕ, we mean a set N such that $\phi(A) = 0$ for every measurable $A \subset N$.)

14. Complete the proof of Theorem 10.44 for $p = 1$.

15. (Converse of Hölder's inequality) Let μ be a σ-finite measure and $1 \leq p \leq \infty$.

(a) Show that

$$\|f\|_p = \sup \left| \int fg \, d\mu \right|,$$

where the supremum is taken over all bounded measurable functions g that vanish outside a set (depending on g) of finite measure, and for which $\|g\|_{p'} \leq 1$ and $\int fg \, d\mu$ exists. (If $1 < p \leq \infty$ and $\|f\|_p < \infty$, this can be deduced from Theorem 10.44.)

(b) Show that a real-valued measurable f belongs to L^p if $fg \in L^1$ for all $g \in L^{p'}$, $1/p + 1/p' = 1$.

16. Consider a convolution operator $Tf(x) = \int_{\mathbb{R}^n} f(y)K(x-y)\,dy$ with $K \geq 0$. If $1 \leq p \leq \infty$ and $\|Tf\|_p \leq M\|f\|_p$ for all f, show that $\|Tf\|_{p'} \leq M\|f\|_{p'}$ for all f, $1/p + 1/p' = 1$. (Use Exercise 15 to write $\|Tf\|_{p'} = \sup_{\|g\|_p \leq 1} |\int_{\mathbb{R}^n} (Tf)g\,dx|$, and note that

$$\int_{\mathbb{R}^n} (Tf)(x)g(x)\,dx = \int_{\mathbb{R}^n} (T\widetilde{g})(-y)f(y)\,dy$$

where $\widetilde{g}(x) = g(-x)$.)

Find a generalization if the hypothesis is instead that $\|Tf\|_q \leq M\|f\|_p$ for all f, where q is a fixed index with $1 \leq q \leq \infty$ and $q \neq p$.

17. Let μ be σ-finite and define $L^p(d\mu)$ to be the class of complex-valued f with $\int |f|^p \, d\mu < +\infty$. Let l be a complex-valued bounded linear functional on $L^p(d\mu)$. If $1 \leq p < \infty$, show that there is a function $g \in L^{p'}(d\mu)$ such that $l(f) = \int fg \, d\mu$. (Here, as usual, we define $\int h \, d\mu = \int h_1 \, d\mu + i \int h_2 \, d\mu$ if $h = h_1 + ih_2$ with h_1 and h_2 real-valued.) (Hint: Reduce to the real case.)

18. Give an example to show that $(L^\infty)'$ cannot be identified with L^1 as in Theorem 10.44. (Consider $L^\infty[-1,1]$ with Lebesgue measure, and let $\mathcal{C}$ be the subspace of continuous functions on $[-1,1]$ with the sup norm. Define $l(f) = f(0)$ for $f \in \mathcal{C}$. Then l is a bounded linear functional on $\mathcal{C}$, so by the Hahn–Banach theorem [See, e.g., Theorem 2.5, p. 33 of M. Schechter, *Principles of Functional Analysis*, 2nd edition, Graduate Studies in Mathematics, vol. 36 (2001), American Mathematical Society; or Theorem 1.1, p. 1, of H. Brezis, *Functional Analysis, Sobolev Spaces and Partial Differential Equations*, Springer, 2011.], l has an extension $l \in (L^\infty[-1,1])'$. If there were a function $g \in L^1[-1,1]$ such that $l(f) = \int_{-1}^1 fg\,dx$ for all $f \in L^\infty[-1,1]$, then we would have $f(0) = \int_{-1}^1 fg\,dx$ for all $f \in \mathcal{C}$. Show that this implies that $g = 0$ a.e., so that $l \equiv 0$. The functional l is called the Dirac δ-function.)

To show that $(L^\infty(E;dx))'$ and $L^1(E;dx)$ are not isometrically isomorphic, one can combine the following three facts: $L^1(E;dx)$ is separable; $L^\infty(E;dx)$ is not separable; and, a Banach space is separable if its dual space is separable. For the latter, see the references in the reference note above, Theorem 8.11 on p. 192 of the first, or Theorem 3.26, p. 73, of the second.

19. Complete the proof of Theorem 10.49.

20. Under the hypothesis of Theorem 10.49, prove that

$$\lim_{h\to 0} \frac{1}{\mu(Q_x(h))} \int_{Q_x(h)} |f(y) - f(x)|\, d\mu(y) = 0 \quad \text{a.e. } (\mu).$$

21. Derive an analogue of the Besicovitch Covering Lemma for the case of two dimensions (x, y) when the squares $Q_{(x,y)}$ are replaced by rectangles $R_{(x,y)}(h)$ centered at (x, y) whose x and y dimensions are h and h^2, respectively. Use this result to prove that under the hypothesis of Theorem 10.49,

$$\lim_{h\to 0} \frac{1}{\mu(R_{(x,y)}(h))} \int_{R_{(x,y)}(h)} f \, d\mu = f(x, y) \quad \text{a.e. } (\mu).$$

22. Let μ be a measure and A be a set with $0 < \mu(A) < \infty$. Let f be measurable and bounded on A, and let ϕ be convex in an interval containing the range of f. Prove that

$$\phi\left(\frac{\int_A f\,d\mu}{\int_A d\mu}\right) \leq \frac{\int_A \phi(f)\,d\mu}{\int_A d\mu}.$$

(This is Jensen's inequality for measures. See Theorem 7.44.)

23. A sequence $\{\phi_k\}$ of set functions is said to be uniformly absolutely continuous with respect to a measure μ if given $\varepsilon > 0$, there exists $\delta > 0$ such that if E satisfies $\mu(E) < \delta$, then $|\phi_k(E)| < \varepsilon$ for all k. If $\{f_k\}$ is a sequence of integrable functions on a finite measure space $(\mathcal{S}, \Sigma, \mu)$ that converges pointwise a.e. (μ) to an integrable f, show that $f_k \to f$ in $L(d\mu)$ norm if and only if the indefinite integrals of the f_k are uniformly absolutely continuous with respect to μ. (Cf. Exercise 17 of Chapter 7.)

24. Let $(\mathcal{S}, \Sigma, \mu)$ be a σ-finite measure space, and let f be Σ-measurable and integrable over $\mathcal{S}$. Let Σ_0 be a σ-algebra satisfying $\Sigma_0 \subset \Sigma$. Of course, f may not be Σ_0-measurable. Show that there is a unique function f_0 that is Σ_0-measurable such that $\int fg\,d\mu = \int f_0 g\,d\mu$ for every Σ_0-measurable g for which the integrals are finite. The function f_0 is called the conditional expectation of f with respect to Σ_0, denoted $f_0 = E(f \mid \Sigma_0)$. (Apply the Radon–Nikodym theorem to the set function $\phi(E) = \int_E f\,d\mu$, $E \in \Sigma_0$.)

25. Using the notation of the preceding exercise, prove the following:

(a) $E(af + bg \mid \Sigma_0) = aE(f \mid \Sigma_0) + bE(g \mid \Sigma_0)$, a, b constants.

(b) $E(f \mid \Sigma_0) \geq 0$ if $f \geq 0$.

(c) $E(fg \mid \Sigma_0) = gE(f \mid \Sigma_0)$ if g is Σ_0-measurable.

(d) If $\Sigma_1 \subset \Sigma_0 \subset \Sigma$, then $E(f \mid \Sigma_1) = E(E(f \mid \Sigma_0) \mid \Sigma_1)$.

26. (Hardy's inequality) Let $f \geq 0$ on $(0,\infty)$, $1 \leq p < \infty$, $d\mu(x) = x^{\alpha}\,dx$ and $d\nu(x) = x^{\alpha+p}\,dx$ on $(0,\infty)$. Prove there exists a constant c independent of f such that

(i) $\displaystyle\int_0^{\infty}\left(\int_0^{x} f(t)\,dt\right)^{p} d\mu(x) \leq c\int_0^{\infty} f^{p}(x)\,d\nu(x)$, $\alpha < -1$,

(ii) $\displaystyle\int_0^{\infty}\left(\int_x^{\infty} f(t)\,dt\right)^{p} d\mu(x) \leq c\int_0^{\infty} f^{p}(x)\,d\nu(x)$, $\alpha > -1$.

(For (i), $(\int_0^x f(t)\,dt)^p \leq cx^{p-\eta-1}\int_0^x f(t)^p t^{\eta}\,dt$ by Hölder's inequality, provided $p-\eta-1>0$. Multiply both sides by x^{α}, integrate over $(0,\infty)$, change the order of integration, and observe that an appropriate η exists since $\alpha < -1$.)

27. If μ is a σ-finite regular Borel measure on $\mathbb{R}^n$, show that the class of continuous functions with compact support is dense in $L^p(d\mu)$, $1 \leq p < \infty$. (By Exercise 8, it is enough to approximate χ_E, where E is a Borel set with finite measure. Given $\varepsilon > 0$, as shown in Section 10.5 on p. 269, there exist open G and closed F with $F \subset E \subset G$ and $\mu(G - F) < \varepsilon$. Now use Urysohn's lemma: if F_1 and F_2 are disjoint closed sets in $\mathbb{R}^n$, there is a continuous f on $\mathbb{R}^n$ with $0 \leq f \leq 1$, $f = 1$ on F_1, $f = 0$ on F_2.)

28. Let $1 < p \leq \infty$ and μ be a σ-finite measure for which $L^{p'}(E; d\mu)$ is separable, $1/p + 1/p' = 1$. Show that every bounded sequence in $L^p(E; d\mu)$ has a weakly convergent subsequence, that is, if $\sup_k \|f_k\|_p < \infty$, show that there exists $\{f_{k_j}\}$ and $f \in L^p$ such that $\int_E f_{k_j} g \, d\mu \to \int_E f g \, d\mu$ for all $g \in L^{p'}$. (Use the Bolzano–Weierstrass theorem to show that for every $g \in L^{p'}$, there is a subsequence $\{f_{k_j}\}$ depending on g such that $\int_E f_{k_j} g \, d\mu$ converges. By using a diagonal argument, $\{f_{k_j}\}$ can be chosen to be independent of g for all g in any fixed countable subset S of $L^{p'}$ and consequently for all $g \in L^{p'}$ by choosing S to be dense in $L^{p'}$. Finally, apply Theorem 10.44 to the linear functional $l \in (L^{p'})'$ defined by $l(g) = \lim_{k_j \to \infty} \int_E f_{k_j} g \, d\mu$.)

29. Let l^p be defined as in Section 8.3. Explain how Theorem 10.44 can be used to describe both the action and the norm of a continuous linear functional on l^p, $1 \leq p < \infty$.

30. Let Σ be the σ-algebra of Lebesgue measurable sets in $\mathbb{R}^1$. For every $E \in \Sigma$, let $\Lambda(E)$ denote the Lebesgue measure of E, and define measures R and Δ by $R(E) = \Lambda(E \cap [0,1])$ and $\Delta(E) = \chi_E(0)$:

(a) Show that $\int_E f\, d\Delta = f(0)\Delta(E)$.

(b) Is either R or Δ absolutely continuous or singular with respect to Λ?

(c) Identify the functions f and the sets Z in the Lebesgue decompositions of Δ with respect to Λ, of Λ with respect to Δ, of R with respect to Λ, and of Λ with respect to R.

31. Prove the Besicovitch Covering Lemma in case $n = 1$.

32. Let $w(x)$ be a nonnegative locally integrable function on $\mathbb{R}^n$ such that $\int_Q w > 0$ for every cube Q in $\mathbb{R}^n$ with edges parallel to the coordinate axes. Consider the w-weighted maximal function $M_w f$ defined by

$$M_w f(x) = \sup \frac{1}{\int_Q w(y)\,dy} \int_Q |f(y)|w(y)\,dy, \qquad x \in \mathbb{R}^n,$$

where f is a measurable function and the supremum is taken over all cubes $Q \subset \mathbb{R}^n$ centered at x with edges parallel to the coordinate axes. Show that

$$\int_{\{x:M_w f(x)>\alpha\}} w(x)\,dx \le \frac{C}{\alpha} \int_{\mathbb{R}^n} |f(x)|w(x)\,dx, \quad \alpha > 0,$$

and that for $1 < p \le \infty$,

$$\left(\int_{\mathbb{R}^n} |M_w f(x)|^p w(x)\,dx \right)^{1/p} \le C \left(\int_{\mathbb{R}^n} |f(x)|^p w(x)\,dx \right)^{1/p},$$

where the constant C is independent of w, f, and α. (Compare Lemma 10.47 and Theorem 9.16.)

11. Outer Measure and Measure

1.

(a) Prove the second statements in both parts of Corollary 11.4.

(b) Verify the statements made before Theorem 11.5 about the function $r(A)$ defined on sets $A \subset \mathbb{R}^2$. (One way to see that a set B with $d(Y, B) > 0$ is not r-measurable is to denote the mirror reflection of B in the y-axis by B^* and check that the equation $r(B \cup B^*) = r(B) + r(B^*)$ is false.)

2. Let μ be a finite Borel measure on $\mathbb{R}^1$, and define $f_\mu(x) = \mu((-\infty, x])$, $-\infty < x < +\infty$. Show that f_μ is monotone increasing, $\mu((a, b]) = f_\mu(b) - f_\mu(a)$, f_μ is continuous from the right, and $\lim_{x\to-\infty} f_\mu(x) = 0$.

3. Let f be monotone increasing on $\mathbb{R}^1$.

(a) Show that $\Lambda_f(\mathbb{R}^1)$ is finite if and only if f is bounded.

(b) Let f be bounded and right continuous, let $\mu = \Lambda_f$, and let $\bar{f}$ denote the function f_μ defined in Exercise 2. Show that f and $\bar{f}$ differ by a constant.

Thus, if we make the additional assumption that $\lim_{x\to-\infty} f(x) = 0$, then $f = \bar{f}$.

4. If we identify two functions on $\mathbb{R}^1$ which differ by a constant, prove that there is a one-to-one correspondence between the class of finite Borel measures on $\mathbb{R}^1$ and the class of bounded increasing functions that are continuous from the right.

5. Let f be monotone increasing and right continuous on $\mathbb{R}^1$.

(a) Show that Λ_f is absolutely continuous with respect to Lebesgue measure if and only if f is absolutely continuous on $\mathbb{R}^1$. (By absolutely continuous on $\mathbb{R}^1$, we mean absolutely continuous on every compact interval.)

(b) If Λ_f is absolutely continuous with respect to Lebesgue measure, show that its Radon–Nikodym derivative equals df/dx.

6. Prove that the Lebesgue–Stieltjes outer measure constructed from $f(x) = x$ is the same as Lebesgue outer measure.

7. If f is monotone increasing and continuous from the right on $\mathbb{R}^1$, show that $\Lambda_f^*(A) = \Lambda_f^{\circ *}(A)$, where $\Lambda_f^{\circ *}$ is defined in the same way as Λ_f^* except that we use open intervals (a_k, b_k).

8. If f is monotone increasing and continuous from the right, derive formulas for $\Lambda_f([a,b])$ and $\Lambda_f((a,b))$.

9. Complete the proof of Theorem 11.12.

10. Show that in $\mathbb{R}^n$, $n > 1$, the Hausdorff outer measure H_n is not identical to Lebesgue outer measure. (For example, let $n = 2$, and write $A = \bigcup A_k$, $\delta(A_k) < \varepsilon$. Enclose A_k in a circle C_k with the same diameter, and show that $\sum \delta(A_k)^2 \geq (4/\pi)|A|_e$. Thus, $H_2^{\varepsilon}(A) \geq (4/\pi)|A|_e$.)

11. If A is a subset of $\mathbb{R}^n$, define the Hausdorff dimension of A as follows: If $H_\alpha(A) = 0$ for all $\alpha > 0$, let $\dim A = 0$; otherwise, let

$$\dim A = \sup\{\alpha : H_\alpha(A) = +\infty\}.$$

(a) Show that $H_\alpha(A) = 0$ if $\alpha > \dim A$ and that $H_\alpha(A) = +\infty$ if $\alpha < \dim A$. Show that in $\mathbb{R}^n$ we have $\dim A \leq n$. See Exercise 19 in order to determine the Hausdorff dimension of the Cantor set.

(b) If $\dim A_k = d$ for each A_k in a countable collection $\{A_k\}$, show that $\dim(\bigcup A_k) = d$. Hence, show that every countable set has Hausdorff dimension 0.

12. Let Γ be an outer measure on $\mathcal{S}$, and let Γ' denote Γ restricted to the Γ-measurable sets. Since Γ' is a measure on an algebra, it induces an outer measure Γ^*. Show that $\Gamma^*(A) \geq \Gamma(A)$ for $A \subset \mathcal{S}$ and that equality holds for a given A if and only if there is a Γ-measurable set E such that $A \subset E$ and $\Gamma(E) = \Gamma(A)$. Thus, $\Gamma = \Gamma^*$ if Γ is regular.

13. Let λ be a measure on an algebra $\mathcal{A}$, and let λ^* be the corresponding outer measure. Given A, show that there is a set H of the form $\bigcap_k \bigcup_j A_{k,j}$ such that $A_{k,j} \in \mathcal{A}$, $A \subset H$ and $\lambda^*(A) = \lambda^*(H)$. Thus, every outer measure that is induced by a measure on an algebra is regular.

14. Prove Theorem 11.18.

15.

(a) Show that the intersection of a family of algebras is an algebra.

(b) A collection $\mathcal{C}$ of subsets of $\mathcal{S}$ is called a subalgebra if it is closed under finite intersections and if the complement of any set in $\mathcal{C}$ is the union of a finite number of disjoint sets in $\mathcal{C}$. Give an example of a subalgebra. Show that a subalgebra $\mathcal{C}$ generates an algebra by adding $\varnothing$, $\mathcal{S}$, and all finite disjoint unions of sets of $\mathcal{C}$.

16. If μ is a finite Borel measure on $\mathbb{R}^1$, show that $\mu(B) = \sup \mu(F)$ for every Borel set B, where the sup is taken over all closed subsets F of B.

17. Show that the conclusions of Theorems 10.48 and 10.49, and therefore also the conclusion of Corollary 10.50, remain true without the assumption (ii) stated before Lemma 10.47. (Show that without this assumption, the conclusions of Lemma 10.47 are true with μ replaced by μ^*; for example,

$$\mu^* \left\{ x \in E : \sup_{h>0} \frac{\nu(Q_x(h))}{\mu(Q_x(h))} > \alpha \right\} \leq c \frac{\nu(\mathbb{R}^n)}{\alpha}.$$

)

18. Derive analogues of Corollary 11.21 and Theorem 11.24 in $\mathbb{R}^n$, $n > 1$. (Use partly open n-dimensional intervals in place of the intervals $(a, b]$.)

19. Show that the Cantor set C in $[0,1]$ has the Hausdorff measure of order $\log 2/\log 3$ equal to 1. (In order to show the inequality $H_{\log 2/\log 3}(C) \le 1$, consider $H^{(\varepsilon)}_{\log 2/\log 3}(C)$ when $\varepsilon = 3^{-k}$, $k = 1, 2, \ldots$, and show that the 2^k intervals $\{I_j\}$ of length 3^{-k} remaining at the kth stage C_k of construction of C satisfy $\sum |I_j|^\alpha = 2^k 3^{-k\alpha} = 1$ if $\alpha = \log 2/\log 3$. A proof of the opposite inequality is harder. It may be helpful to note that if I is a closed interval, containing at its two endpoints intervals from C_k, then $|I|^\alpha \ge n_k(I)/2^k$, where $n_k(I)$ is the number of intervals of C_k contained in I and $\alpha = \log 2/\log 3$.)

12. A Few Facts from Harmonic Analysis

1. Prove the following versions of Theorems 12.13 and 12.14.

(a) If $f \sim \frac{1}{2}a_0 + \sum_{k=1}^{\infty}(a_k \cos kx + b_k \sin kx)$, and f is the indefinite integral of f', then $f' \sim \sum_{k=1}^{\infty} k(b_k \cos kx - a_k \sin kx)$.

(b) If $f \sim \frac{1}{2}a_0 + \sum_{k=1}^{\infty}(a_k \cos kx + b_k \sin kx)$, and F is the indefinite integral of f, then

$$F(x) - \tfrac{1}{2}a_0 x \sim \tfrac{1}{2}A_0 + \sum_{k=1}^{\infty} \frac{1}{k}(a_k \sin kx - b_k \cos kx).$$

2. Prove that if $f(x) \sim \sum c_k e^{ikx}$, then $f(x+\alpha) \sim \sum c_k e^{ik\alpha} e^{ikx}$.

3. If f is real or complex-valued and $f \sim \frac{1}{2}a_0 + \sum_{k=1}^{\infty}(a_k \cos kx + b_k \sin kx)$, show that

$$\frac{1}{\pi}\int_0^{2\pi} |f|^2 = \frac{1}{2}|a_0|^2 + \sum_{k=1}^{\infty}(|a_k|^2 + |b_k|^2).$$

4. Deduce from (12.18) that if $f, g \in L^2$, $f \sim \sum c_k e^{ikx}$, $g \sim \sum d_k e^{ikx}$, then

$$\frac{1}{2\pi}\int_0^{2\pi} f(t)g(x-t)\,dt = \sum c_k d_k e^{ikx}.$$

Thus, a Fourier coefficient of the convolution of f and g equals the product of the corresponding coefficients of f and g. (For nonperiodic versions of this result, see Theorems 13.30 and 13.59.)

5. Prove the following.

(a) If $f(x)$ is periodic and equal to $\operatorname{sign} x$ in $(-\pi, \pi)$, then

$$f(x) \sim \frac{4}{\pi}\left(\sin x + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \cdots\right).$$

(b) Let $0 < h < \frac{1}{2}\pi$, and let f be the triangular function defined as follows: f is periodic, even, continuous, $f(0) = 1$, $f(x) = 0$ for $2h \le x \le \pi$, and f is linear in $(0, 2h)$. Then

$$\begin{aligned} f &\sim \frac{2h}{\pi}\left[\frac{1}{2} + \sum_{k=1}^{\infty}\left(\frac{\sin kh}{kh}\right)^2 \cos kx\right] \\ &= \frac{h}{\pi}\left[1 + \sum_{k=-\infty}^{+\infty}{}' \left(\frac{\sin kh}{kh}\right)^2 e^{ikx}\right]. \end{aligned}$$

(c) Let g be periodic and equal to $\frac{1}{2}\log(1/|2\sin\frac{1}{2}x|)$ in $(-\pi, \pi)$. Then

$$g \sim \sum_{k=1}^{\infty} \frac{\cos kx}{k}.$$

(For (b), the coefficients can be computed directly, or by using Exercise 4 and Example (b), Section 12.1, p. 306. Observe that the convolution of the characteristic function of an interval with itself is a triangular function. For (c), one may either integrate by parts in the formula for the cosine coefficients of g or consider the real part of the series

$$\sum_{k=1}^{\infty} \frac{z^k}{k} = \log\frac{1}{1-z}, \qquad z = re^{ix},$$

for $r < 1$, and then let $r \to 1$.)

6. Using the formula for the Fourier series of $\frac{1}{2}(\pi - x)$ given in Example (a), Section 12.1, p. 306, prove the formulas

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}, \qquad \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90},$$

$$\sum_{n=1}^{\infty} \frac{1}{n^{2k}} = \pi^{2k} B_k \quad \text{for some rational } B_k \quad (k = 1, 2, \ldots).$$

7. Prove that each of the systems

(a) $\frac{1}{2}, \cos x, \cos 2x, \ldots, \cos kx, \ldots$

(b) $\sin x, \sin 2x, \ldots, \sin kx, \ldots$

is orthogonal and complete over $(0, \pi)$.

8. Let $\{\phi_j(x)\}$ and $\{\psi_k(y)\}$ be two orthogonal systems: the first over a set $A \subset \mathbb{R}^m$ and the second over a set $B \subset \mathbb{R}^n$. Then the (double) system

$$\omega_{j,k}(x,y) = \phi_j(x)\psi_k(y)$$

is orthogonal over the Cartesian product $C = A \times B \subset \mathbb{R}^{m+n}$. If both $\{\phi_j\}$ and $\{\psi_k\}$ are complete, so is $\{\omega_{j,k}\}$.

Generalize this to the case of more than two orthogonal systems.

9. Let $\{(k_1, k_1, \ldots, k_n)\}$ be all lattice points in the space $\mathbb{R}^n$ (i.e., all distinct points with integral coordinates). Then the system

$$\exp\{i(k_1x_1 + k_2x_2 + \cdots + k_nx_n)\}$$

is orthogonal and complete over any cube in $\mathbb{R}^n$ with edge length 2π.

10. If $f \sim \sum c_k e^{ikx} \in L^2$ and $g \sim \sum d_k e^{ikx} \in L^2$, then $h = fg$ is integrable, and if $h \sim \sum C_n e^{inx}$, then

$$C_n = \sum_{k=-\infty}^{+\infty} c_k d_{n-k},$$

where the series on the right converges absolutely. (For $n = 0$, this means $(1/2\pi) \int_0^{2\pi} fg = \sum c_k d_{-k}$, which is a variant of (12.18).) See also Exercises 17 and 18.

11. Let $f \sim \sum c_k e^{ikx} \in L^2$. For each n, let

$$\gamma_n = \sum_{k \neq n} c_k \frac{1}{n-k} = \sum_{k \neq 0} c_{n-k} \frac{1}{k}.$$

Show that $\{\gamma_n\} \in l^2$ and, more precisely, that $\sum |\gamma_n|^2 \leq \pi^2 \sum |c_k|^2$.

The numbers γ_n are discrete analogues of the formal Hilbert integral $\int [f(t)/(x-t)]\, dt$; see Section 3 of Chapter 13. (The numbers γ_n are the Fourier coefficients of the function fg, where $g \sim \sum_{k \neq 0} e^{ikx}/k = i(\pi - x)$ for $0 < x < 2\pi$ (see Example (a), Section 12.1, p. 306), and we have

$$\int_0^{2\pi} |fg|^2 \leq \pi^2 \int_0^{2\pi} |f|^2.$$

)

12. Let f and g be periodic, $f \in L^p$, $g \in L^{p'}$, $1 \le p \le \infty$, $1/p + 1/p' = 1$. Consider the product $h_x(t) = f(x+t)g(t)$ as a function of t. Show that the nth Fourier coefficient of $h_x(t)$ tends to 0 as $n \to \infty$, uniformly in x. (Show that the L^1-modulus of continuity of h tends to 0 uniformly in x; apply (12.22).)

13. Let f be periodic and integrable. Show that for the partial sums s_n and $\widetilde{s}_n$, we have the following formulas:

(a) $s_n(x) = (1/\pi)\int_{-\pi}^{\pi} f(x+t)[(\sin nt)/t]\,dt + \varepsilon_n(x)$, where $\varepsilon_n(x)$ tends to 0 uniformly in x as $n \to \infty$.

(b) $\widetilde{s}_n(x) = (1/\pi)\int_{-\pi}^{\pi} f(x+t)[(1-\cos nt)/t]\,dt + \eta_n(x)$, where $\eta_n(x)$ tends to 0 uniformly in x.

(For (a), except for an error which is $o(1)$ uniformly in x, we have

$$s_n(x) = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x+t)\frac{\sin nt}{2\tan\frac{1}{2}t}\,dt,$$

and since $\frac{1}{2}\cot\frac{1}{2}t - \frac{1}{t}$ is bounded in $(-\pi,\pi)$, the result follows easily from Exercise 12 with $p=1$, $p'=\infty$.)

14. Let $\widetilde{L}_n = 2\pi^{-1}\int_0^\pi |\widetilde{D}_n(t)|\,dt$, $\widetilde{D}_n$ denoting the conjugate Dirichlet kernel. Prove the following analogue of Theorem 12.36:

$$\widetilde{L}_n = \frac{2}{\pi}\log n + o(\log n) \quad \text{as } n \to \infty.$$

Show also that $\widetilde{L}_n = |\widetilde{s}_n(0, \operatorname{sign}\widetilde{D}_n)| = \max\{|\widetilde{s}_n(0,f)| : f \text{ with } |f| \le 1\}$.

15. Show that if x is a point of continuity of an integrable f, then $\widetilde{s}_n(x) = o(\log n)$. If f has a jump discontinuity at x and the value of the jump is $d = f(x+) - f(x-)$, show that $\widetilde{s}_n(x) = -(d/\pi)\log n + o(\log n)$. (The second statement follows from the first by considering [if, e.g., $x = 0$] the function $h(t) = \frac{1}{2}(\pi - t)$ of Example (a), Section 12.1, p. 306, whose conjugate function $\widetilde{h}$ satisfies $\widetilde{h} = -g$, where g is the function of Exercise 5(c).)

16. Prove the following more general form of Theorem 12.38. Suppose that $\liminf s_n = \underline{s}$ and $\limsup s_n = \bar{s}$ are finite. Then both $\liminf \sigma_n$ and $\limsup \sigma_n$ are contained between the numbers

$$\tfrac{1}{2}(\underline{s} + \bar{s}) \pm A\tfrac{1}{2}(\bar{s} - \underline{s}),$$

where A is the same as in condition (i) of Theorem 12.38.

17. Prove the following extension of Exercise 10. Let $f \in L^p$ and $g \in L^{p'}$, $1 < p < \infty$, $1/p+1/p' = 1$ (thus also $1 < p' < \infty$). If $f \sim \sum c_k e^{ikx}$ and $g \sim \sum d_k e^{ikx}$, then the Fourier coefficients C_n of the (integrable) function $h = fg$ are given by

$$C_n = \sum_{k=-\infty}^{\infty} c_k d_{n-k} = \lim_{M\to+\infty} \sum_{k=-M}^{M} c_k d_{n-k}.$$

Thus, the C_n are the same as in Exercise 10, but the series representing the C_n are no longer claimed to be absolutely convergent, and we must consider the limits of their symmetric partial sums. (The proof is parallel to that of Exercise 10. We write

$$\begin{aligned} C_n &= \frac{1}{2\pi}\int_0^{2\pi} fge^{-inx} \\ &= \frac{1}{2\pi}\int_0^{2\pi} (f - s_M)ge^{-inx} + \frac{1}{2\pi}\int_0^{2\pi} s_M ge^{inx}, \end{aligned}$$

where $s_M = s_M(x, f)$, observe that the first integral on the right is majorized by $\int_0^{2\pi} |f - s_M||g|$, apply Hölder's inequality, and use the fact that $\|f - s_M\|_p \to 0$ by Theorem 12.88.)

18. The result of Exercise 17 is valid when $p = 1$, $p' = \infty$ (or when $p = \infty$, $p' = 1$), but the series defining the C_n must be taken in the sense of the (symmetric) first arithmetic means:

$$C_n = \lim_{M \to +\infty} \sum_{k=-M}^{M} c_k \left(1 - \frac{|k|}{M+1}\right) d_{n-k}.$$

19. We know that Theorem 12.79 is false for $p = 1$. Show that it is also false for $p = \infty$, that is, that the conjugate function of a bounded function need not be bounded. (Consider, e.g., the two series $\sum_{k=1}^{\infty}(\sin kx)/k \sim \frac{1}{2}(\pi - x)$, $0 < x < 2\pi$, and $\sum_{k=1}^{\infty}(\cos kx)/k \sim \frac{1}{2}\log(1/|2\sin\frac{1}{2}x|)$, $-\pi < x < \pi$ [see Exercise 5].)

20. There is a substitute result for Theorem 12.79 in case $p = \infty$. Let f be a periodic function with $|f| \le 1$. Then there are absolute constants $\lambda, \mu > 0$ such that

$$\int_{-\pi}^{\pi} e^{\lambda|\widetilde{f}|} \le \mu.$$

See also Exercise 27 in Chapter 14. (Write

$$e^{\lambda|\widetilde{f}|} - \lambda|\widetilde{f}| - 1 = \sum_{n=2}^{\infty} \frac{\lambda^n |\widetilde{f}|^n}{n!}$$

and integrate termwise. Use (12.80)(a), (12.85), and the fact that $n^n \le c^n n!$.)

21. Suppose that f is a periodic function for which

$$\int_{-\pi}^{\pi} |f| \log^+ |f| < +\infty,$$

where $\log^+$ stands for the positive part of log. This clearly implies that $f \in L^1$. Show that $\widetilde{f} \in L^1$ and that there are absolute constants A and B such that

$$\int_{-\pi}^{\pi} |\widetilde{f}| \le A \int_{-\pi}^{\pi} |f| \log^+ |f| \, dx + B.$$

(Write $\omega(\alpha) = |\{x : |x| < \pi,\ |\widetilde{f}(x)| > \alpha\}|$. Then

$$\int_{-\pi}^{\pi} |\widetilde{f}| = \int_0^{\infty} \omega(\alpha) \, d\alpha = \int_0^2 + \int_2^{\infty}.$$

For the first integral on the right, use the fact that $\omega(\alpha) \le 2\pi$, and for the second, use an argument like that in the proof of Lemma 12.81.)

22. The discussion that precedes Theorem 12.31 shows only that the averages $(\widetilde{s}_n(x) + \widetilde{s}_{n-1}(x))/2$ converge. Give the remaining details of the proof of Theorem 12.31. (Consider $\widetilde{s}_n(x) - \widetilde{s}_{n-1}(x)$.)

23.

(a) Prove the statement about nontangential approach made before Theorem 12.42, namely, that given $\gamma > 0$, there exist $C, \delta > 0$ with $\delta < 1$ such that $|1 - z| \leq C(1 - |z|)$ if $z = x + iy$, $|z| < 1$, $1 - \delta < x < 1$ and $|y| < \gamma(1 - x)$. (It may be helpful to use the asymptotic estimates $|\sin\theta| \sim |\theta|$ and $1 - \cos\theta \sim \theta^2/2$ as $\theta \to 0$.)

(b) Verify that (12.65) characterizes the nontangential approach of $z = re^{ix}$ to 1, $|z| < 1$.

13. The Fourier Transform

1. Let $f \in L^1(\mathbb{R}^n)$. Show that if f is real-valued and even, then $\widehat{f}$ is also real-valued and even. If f is real-valued and odd, show that $\widehat{f}$ is odd and purely imaginary.

Find a complex-valued f such that $\widehat{f}(x)$ is real-valued and finite for all x.

2. Let f be an even integrable function on $(-\infty, \infty)$. Show that

$$\widehat{f}(x) = \frac{1}{\pi} \int_0^\infty f(y) \cos(|x|y)\, dy.$$

3. Let $f(x) \in L^1(\mathbb{R}^n)$, $n \geq 2$, and suppose that f is a radial function: $f(x) = g(|x|)$ for all x. Assume that the polar coordinate representation

$$\int_{\mathbb{R}^n} f(x)\,dx = \int_0^\infty \left(\int_{|x'|=1} f(rx')\,dx' \right) r^{n-1}\,dr$$

is valid, where dx' denotes the differential element of surface area on the unit sphere $\{x' \in \mathbb{R}^n : |x'| = 1\}$, and $x = rx'$, $r = |x|$ if $|x| \neq 0$. Show that

$$\widehat{f}(x) = (2\pi)^{-n/2}|x|^{-(n-2)/2} \int_0^\infty g(r) r^{n/2} J_{(n-2)/2}(r|x|)\,dr,$$

where $J_{(n-2)/2}(t)$, $t > 0$, is the Bessel function of order $(n-2)/2$, which may be defined as follows. For $t \geq 0$ and x' with $|x'| = 1$, show that the function of (t, x') defined by $\int_{|y'|=1} e^{-itx'\cdot y'}\,dy'$ is independent of x', and then set

$$\int_{|y'|=1} e^{-itx'\cdot y'}\,dy' = (2\pi)^{n/2} t^{-(n-2)/2} J_{(n-2)/2}(t), \quad t > 0.$$

Note in particular that the formula above for $\widehat{f}$ shows that $\widehat{f}$ is a radial function if f is radial.

4. Verify properties (13.5) and (13.6) of the Fourier transform.

5. Let L be a nonsingular linear transformation of $\mathbb{R}^n$ and define $(Lf)(x) = f(Lx)$. If $f \in L^1(\mathbb{R}^n)$, show that

$$\widehat{Lf}(x) = |\det L|^{-1} \widehat{f}((L^{-1})^* x), \qquad x \in \mathbb{R}^n,$$

where $(L^{-1})^*$ $(= (L^*)^{-1})$ is the adjoint (or transpose) of L^{-1}. Equivalently, $\widehat{(Lf)} = |\det L|^{-1} \widetilde{L} \widehat{f}$ where $\widetilde{L} = (L^{-1})^*$.

6.

(a) Verify (13.12). Also, given a function $f \in L^1(\mathbb{R}^n)$, find a function $f(x,t)$ defined on $\mathbb{R}^{n+1}_+$ such that

$$\sum_{k=1}^{n} \frac{\partial^2}{\partial x_k^2} f(x,t) = \frac{\partial}{\partial t} f(x,t) \quad \text{in } \mathbb{R}^{n+1}_+$$

and

$$\lim_{t \to 0} f(x,t) = f(x) \quad \text{at every Lebesgue point } x \text{ of } f.$$

Such a function $f(x,t)$ is said to solve the Dirichlet problem for the heat equation in $\mathbb{R}^{n+1}_+$. (Consider dilations of the Gauss–Weierstrass integral $Wf(x,t)$.)

(b) Let $Pf(x,\varepsilon)$ be the Poisson integral of f defined by (13.27) for $(x,\varepsilon) \in \mathbb{R}^{n+1}_+$. If $f \in L^1(\mathbb{R}^n)$, show that Pf is harmonic in $\mathbb{R}^{n+1}_+$, that is, Pf satisfies Laplace's equation

$$\left(\frac{\partial^2}{\partial x_1^2} + \cdots + \frac{\partial^2}{\partial x_n^2} + \frac{\partial^2}{\partial \varepsilon^2} \right) Pf(x,\varepsilon) = 0 \quad \text{if } x \in \mathbb{R}^{n+1}_+,\ \varepsilon > 0.$$

Hence, since $\lim_{\varepsilon \to 0} Pf(x,\varepsilon) = f(x)$ at every Lebesgue point x of f, Pf solves the Dirichlet problem for Laplace's equation in the half-space $\mathbb{R}^{n+1}_+$.

7. Let $f \in L^1(\mathbb{R}^n)$, $P(x,\varepsilon) = P_\varepsilon(x)$ be the Poisson kernel for $\mathbb{R}^{n+1}_+$, and $Pf(x,\varepsilon)$ be the Poisson integral of f. Show that for all $x \in \mathbb{R}^n$ and all $\varepsilon, \delta > 0$,

$$(P_\varepsilon * P_\delta)(x) = P_{\varepsilon+\delta}(x) \quad \text{and} \quad (Pf(\cdot,\varepsilon) * P_\delta)(x) = Pf(x, \varepsilon+\delta).$$

Prove analogues for $W(x,t)$ and $Wf(x,t)$.

8. Prove Theorem 13.40. Note in particular that if $\Delta = \frac{\partial^2}{\partial x_1^2} + \cdots + \frac{\partial^2}{\partial x_n^2}$ denotes the Laplace differential operator (or Laplacian) in $\mathbb{R}^n$, then

$$\widehat{\Delta f}(x) = -|x|^2 \widehat{f}(x) \quad \text{if } f \in \mathcal{S}.$$

(Use Theorem 13.31 and the fact that if $f \in \mathcal{S}$ then $p(x)f(x) \in \mathcal{S}$ for every polynomial $p(x)$.)

9. Let $f \in L^1(\mathbb{R}^n)$. If $\widehat{f} \in \mathcal{S}$, show that there is a function $F \in \mathcal{S}$ such that $f = F$ in the Lebesgue set of f.

10. If $f, g \in \mathcal{S}$, prove that $f * g \in \mathcal{S}$.

11. Suppose that $f \in L^1(\mathbb{R}^n)$ and f has a partial derivative in the L^1 sense with respect to x_k for every $k = 1, \ldots, n$. Prove that $\widehat{f}(x) = o(|x|^{-1})$ as $|x| \to \infty$. Compare the estimate preceding Corollary 13.38 with $N = 1$ there.

12.

(a) Show that the principal value Fourier transform in $\mathbb{R}^1$ of $1/|x|$ is identically infinite.

(b) Show that the principal value Fourier transform in $\mathbb{R}^1$ of $1/|x|^{1-\alpha}$, $0 < \alpha < 1$, equals $c_\alpha |x|^{-\alpha}$ if $x \neq 0$. (We will see in Chapter 14 that the function $1/|x|^{1-\alpha}$, $0 < \alpha < 1$, is the kernel of the one-dimensional fractional integral operator of order α.)

13. Let $1 \leq p \leq \infty$, $1/p + 1/p' = 1$, $f \in L^p(\mathbb{R}^n)$, and $g \in L^{p'}(\mathbb{R}^n)$. Show that $f * g$ is uniformly continuous and bounded on $\mathbb{R}^n$.

14. Part of Plancherel's theorem states that the range of the map $\mathcal{F} : f \to \widehat{f}$, $f \in L^2(\mathbb{R}^n)$, is all of $L^2(\mathbb{R}^n)$, that is, that $\mathcal{F}$ is surjective. Show that this can also be proved as follows without using Corollary 13.38. The range space $\mathcal{F}(L^2(\mathbb{R}^n))$ is a closed subset of $L^2(\mathbb{R}^n)$. If it were a proper subset, there would exist (by using an orthogonal basis in $L^2(\mathbb{R}^n)$) a nontrivial $g \in L^2(\mathbb{R}^n)$ such that $\int_{\mathbb{R}^n} g\widehat{f}\, dx = 0$ for all $f \in L^2(\mathbb{R}^n)$. Now use (13.53) to deduce a contradiction.

15. Verify Theorem 13.55.

16. Let $f \in L^2(-\infty, \infty)$. Show that the singly truncated Hilbert transform

$$H_\varepsilon f(x) = \frac{1}{\pi} \int_{|y|>\varepsilon} f(x-y)\frac{1}{y}\, dy, \quad \varepsilon > 0,\ x \in (-\infty, \infty),$$

is finite and continuous at every x and that $\|H_\varepsilon f - Hf\|_2 \to 0$ as $\varepsilon \to 0$. Show also that there is a constant c independent of f and ε such that $\|H_\varepsilon f\|_2 \le c\|f\|_2$.

17. Given a real-valued function $f \in L^2(\mathbb{R}^1)$ (or more generally in $L^p(\mathbb{R}^1)$ for some p, $1 \le p < \infty$), consider the Cauchy integral $Cf(z)$, $z = x + iy$, defined in the upper half-space by

$$Cf(z) = \frac{1}{i\pi} \int_{-\infty}^{\infty} f(t) \frac{1}{t - z}\, dt, \quad z = x + iy,\ y > 0.$$

(a) Show that for every such z, $Cf(z)$ converges absolutely, has real part equal to the Poisson integral of f, that is, equal to

$$f(x, y) = (f * P_y)(x) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(t) \frac{y}{(x - t)^2 + y^2}\, dt,$$

and has imaginary part

$$\widetilde{f}(x, y) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(t) \frac{x - t}{(x - t)^2 + y^2}\, dt.$$

The latter expression $\widetilde{f}(x, y)$ is called the conjugate Poisson integral of f in the upper half-space $\mathbb{R}^2_+$. Recall that the numerical factor $1/\pi$ is needed in order to guarantee that $\int_{-\infty}^{\infty} P_y(x)\, dx = 1$. In this way, the factor $1/\pi$ is also natural in the definition of $\widetilde{f}(x, y)$ even though its kernel $\widetilde{P}(x) = (1/\pi)x/(x^2 + 1)$ is not integrable at infinity.

(b) Show that at every Lebesgue point x of f, the conjugate Poisson integral $\widetilde{f}(x, y)$ is equiconvergent as $y \to 0+$ with the singly truncated Hilbert transform $H_y f(x)$ defined in Exercise 16, that is, $\widetilde{f}(x, y) - H_y f(x) \to 0$ as $y \to 0+$ if x is a Lebesgue point of f. Compare Theorem 12.51 (ii). (Define

$$k(x) = \widetilde{P}(x) - \frac{1}{\pi} \frac{\chi_{\{|x|>1\}}(x)}{x},$$

where $\widetilde{P}(x)$ is as in part (a), and note that $k(x)$ is bounded with $k(x) = O(|x|^{-3})$ as $|x| \to \infty$. Also, $\int_{-\infty}^{\infty} k(x)\, dx = 0$. Now compute the dilated kernels $k_y(x) = y^{-1}k(x/y)$, $y > 0$, and use the ideas of Theorem 9.13 and Exercise 11 of Chapter 9.)

18. Let $f \in \mathcal{S}(-\infty, \infty)$ and let Hf denote the Hilbert transform of f. If $\int_{-\infty}^{\infty} f(x)\,dx \neq 0$, show that there is no function $g \in \mathcal{S}(-\infty, \infty)$ such that $Hf = g$ a.e.

19. If $f, g \in L^2(-\infty, \infty)$ and Hf, Hg denote their Hilbert transforms, show that

$$\int_{-\infty}^{\infty} (Hf)(Hg)\,dx = \int_{-\infty}^{\infty} fg\,dx \quad \text{and} \quad \int_{-\infty}^{\infty} f(Hg)\,dx = -\int_{-\infty}^{\infty} (Hf)g\,dx.$$

20. Let $-\infty < a < b < \infty$ and $\chi_{(a,b)}(x)$ denote the characteristic function of the interval (a, b). Prove that the truncated Hilbert transform $H_{\varepsilon,\omega}\chi_{(a,b)}$ satisfies

$$\lim_{\substack{\varepsilon\to 0+\\ \omega\to\infty}} (H_{\varepsilon,\omega}\chi_{(a,b)})(x) = \frac{1}{\pi}\log\left|\frac{a-x}{b-x}\right|, \qquad x \neq a, b,$$

where the convergence is in the pointwise sense. Deduce that the Hilbert transform of a function that is bounded and has compact support may not be bounded or have compact support.

21. Let $-\infty < a < b < \infty$ and $\chi_{(a,b)}(x)$ denote the characteristic function of the interval (a, b). Set $c = (a + b)/2$ and $\ell = (b - a)/2$. Show that

$$\widehat{\chi_{(a,b)}}(x) = \frac{1}{\pi}\frac{\sin \ell x}{x}e^{-icx}, \qquad x \in (-\infty, \infty),$$

where the right side should be interpreted as ℓ/π when $x = 0$. Note that $\widehat{\chi_{(a,b)}}$ is not integrable (at infinity). Also, given $h > 0$, sketch the graph of the convolution $\frac{1}{2h}\chi_{(-h,h)} * \chi_{(-h,h)}$ and show that the Fourier transform of this function is the integrable function $(\sin^2 hx)/(\pi h x^2)$ (cf. Exercise 5(b) of Chapter 12).

These examples further illustrate the maxim first observed on p. 387 in Section 13.1 that the smoother a compactly supported function is, the smaller its Fourier transform is at infinity.

22. Let $1 < p < 2$ and $f \in L^p(\mathbb{R}^n)$. Show that the functions

$$\widehat{f}_k(x) = \frac{1}{(2\pi)^n} \int_{|y|<k} f(y) e^{-ix\cdot y}\, dy, \qquad k > 0,$$

converge in $L^{p'}(\mathbb{R}^n)$ to $\widehat{f}$ as $k \to \infty$, $1/p + 1/p' = 1$.

23. Let $1 < p < 2$, $f \in L^1(\mathbb{R}^n)$ and $g \in L^p(\mathbb{R}^n)$. Show that $\widehat{f * g} = (2\pi)^n \widehat{f}\widehat{g}$ a.e.

24. Let $f \in L^p(0, 2\pi)$, $1 < p < 2$, be periodic with period 2π, and let $c_k = (1/2\pi) \int_0^{2\pi} f(t) e^{-ikt}\, dt$, $k = 0, \pm 1, \ldots$, be its Fourier coefficients. Prove that there is a constant C independent of f such that

$$\left(\sum_k |c_k|^{p'} \right)^{1/p'} \leq C \left(\int_0^{2\pi} |f|^p \, dt \right)^{1/p}, \qquad \frac{1}{p} + \frac{1}{p'} = 1.$$

This is the periodic Hausdorff–Young theorem. (Write

$$\begin{aligned} \sum_{k=-\infty}^{\infty} |c_k|^{p'} &= \sum_{j=-\infty}^{\infty} \sum_{\substack{k \\ 2^{j-1} < |c_k| \leq 2^j}} |c_k|^{p'} \\ &\approx \sum_{j=-\infty}^{\infty} 2^{jp'} \sum_{\substack{k \\ 2^{j-1} < |c_k| \leq 2^j}} 1 \end{aligned}$$

in analogy with the first equality in Theorem 5.51. After summing by parts, argue similarly to the proof of Theorem 13.71 with 2^{j-1} playing the role of α. This method will not produce the optimal value of the constant C.)

25. Let $f \in L^p(-\infty, \infty)$, $1 \le p \le 2$. Derive the following pointwise formula for $\widehat{f}$:

$$\widehat{f}(x) = \frac{d}{dx}\left(\frac{1}{2\pi}\int_{-\infty}^{\infty} f(y)\frac{e^{-ixy}-1}{-iy}\,dy\right) \quad \text{a.e.}$$

(Define $F(x) = \int_0^x \widehat{f}$ and $F_N(x) = \int_0^x \widehat{f}_N$ where $f_N = f\chi_{(-N,N)}$, $N > 0$. By Hölder's inequality, $|F(x) - F_N(x)| \le |x|^{1/p}\|\widehat{f} - \widehat{f}_N\|_{p'} \to 0$ as $N \to \infty$. Then $\widehat{f}(x) = F'(x) = (d/dx)\lim_{N\to\infty} F_N(x)$ a.e. Also,

$$F_N(x) = \frac{1}{2\pi}\int_{-N}^{N} f(y)\frac{e^{-ixy}-1}{-iy}\,dy.$$

)

26. Let μ be a finite Borel measure on $\mathbb{R}^n$. Define the Fourier transform of μ by

$$\widehat{\mu}(x) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} e^{-ix\cdot y}\, d\mu(y), \qquad x \in \mathbb{R}^n.$$

(a) Show that $\widehat{\mu}$ is bounded and continuous on $\mathbb{R}^n$ and satisfies $\overline{\widehat{\mu}(-x)} = \widehat{\mu}(x)$ for all x.

(b) If x_0 is a point in $\mathbb{R}^n$ and $\{E\}$ is the class of Borel sets in $\mathbb{R}^n$, define the delta measure δ_{x_0} at x_0 by $\delta_{x_0}(E) = 1$ if $x_0 \in E$ and $\delta_{x_0}(E) = 0$ if $x_0 \notin E$. Show that

$$\widehat{\delta_{x_0}}(x) = \frac{1}{(2\pi)^n} e^{-ix_0\cdot x}.$$

Note that $\widehat{\delta_0}(x)$ is identically equal to the constant $1/(2\pi)^n$.

(c) Let $m = 1, 2, \ldots$, $\{x_j\}_{j=1}^m \subset \mathbb{R}^n$, and $\{z_j\}_{j=1}^m$ be complex numbers. Show that

$$\sum_{j=1}^m \sum_{k=1}^m \widehat{\mu}(x_j - x_k) z_j \overline{z_k} = (2\pi)^n \int_{\mathbb{R}^n} \left| \sum_{j=1}^m z_j \widehat{\delta_{x_j}}(y) \right|^2 d\mu(y).$$

In particular, the sum on the left side is nonnegative. (Conversely, it is known that any bounded continuous complex-valued function g on $\mathbb{R}^n$ that satisfies $\overline{g(-x)} = g(x)$ for all x is the Fourier transform of a finite Borel measure provided g is nonnegative definite in the following sense:

$$\sum_{j=1}^m \sum_{k=1}^m g(x_j - x_k) z_j \overline{z_k} \geq 0$$

for all such m, $\{x_j\}_{j=1}^m$ and $\{z_j\}_{j=1}^m$. See S. Bochner, *Lectures on Fourier Integrals*, Annals of Math. Studies 42, Princeton Univ. Press, 1959.)

27.

(a) If $f \in L^1((0,\infty); dt/t)$, that is, if $\int_0^\infty |f(t)|\, dt/t < \infty$, define the Mellin transform $m(f)$ of f by

$$m(f)(x) = \frac{1}{2\pi} \int_0^\infty f(t) t^{-ix} \frac{dt}{t}, \qquad -\infty < x < \infty.$$

If $f, g \in L^1((0,\infty); dt/t)$, define an analogue $f\#g$ of ordinary convolution by

$$(f\#g)(s) = \int_0^\infty f(st^{-1}) g(t) \frac{dt}{t}, \qquad 0 < s < \infty.$$

Show that $f\#g \in L^1((0,\infty); dt/t)$ and

$$m(f\#g)(x) = m(f)(x)\, m(g)(x), \qquad x \in \mathbb{R}^1.$$

(b) For any f defined on $(0,\infty)$, let $(Ef)(x) = f(e^x)$, $x \in \mathbb{R}^1$. If $f \in L^1((0,\infty); dt/t)$, show that $Ef \in L^1(\mathbb{R}^1)$ and $m(f)(x) = \widehat{Ef}(x)$. (Exercise 11 of Chapter 7 may be helpful.) If $f \in L^2((0,\infty); dt/t)$, explain why $Ef \in L^2(\mathbb{R}^1)$, and define $m(f)$ by $m(f)(x) = \widehat{Ef}(x)$, $x \in \mathbb{R}^1$. Show that

$$\frac{1}{2\pi} \int_{e^{-k}}^{e^k} f(t) t^{-ix} \frac{dt}{t} \to m(f)(x) \quad \text{in } L^2(\mathbb{R}^1) \text{ norm as } k \to \infty,$$

and

$$\int_{-\infty}^\infty |m(f)(x)|^2 dx = \frac{1}{2\pi} \int_0^\infty |f(t)|^2 \frac{dt}{t}.$$

28. Show that $f \equiv 0$ if $f, \widehat{f} \in C_0^\infty(\mathbb{R}^n)$. (In case $n = 1$ and f is supported in the open interval $(-\pi, \pi)$, the Fourier series $S[f]$ of f in $(-\pi, \pi)$ converges everywhere in $(-\pi, \pi)$ to f. If $\widehat{f}$ has compact support, $S[f]$ reduces to a trigonometric polynomial and therefore has at most a finite number of roots (mod 2π); see Section 12.1, p. 307. An alternate proof based on the identity theorem for analytic functions is possible.)

29. There is an analogue for smooth truncations of the limit inversion fact in (13.57). For $k = 1, 2, \ldots$, let $\chi_k(x) \in C_0^\infty(\mathbb{R}^n)$ with $\chi_k(x) = 1$ if $|x| \le k$, $\chi_k(x) = 0$ if $|x| \ge 2k$, and $0 \le \chi_k \le 1$. The operators

$$\Psi_k f(x) = \int_{\mathbb{R}^n} \chi_k(y) \widehat{f}(y) e^{ix\cdot y}\, dy, \qquad f \in L^2(\mathbb{R}^n),$$

are often called pseudodifferential cutoff operators. Show that $\|\Psi_k f\|_2 \le \|f\|_2$ and $\|\Psi_k f - f\|_2 \to 0$ as $k \to \infty$. Show also that $\Psi_k f = \psi_k * f$, where $(2\pi)^n \widehat{\psi_k} = \chi_k$. Deduce that $\Psi_k f$ is infinitely differentiable if $f \in L^2(\mathbb{R}^n)$, and that if $f \in \mathcal{S}$, then $D^\alpha \Psi_k f \to D^\alpha f$ in $L^2(\mathbb{R}^n)$ for every multi-index α.

14. Fractional Integration

1. Show that if $f \in C^1(\overline{B})$, where $\overline{B}$ denotes the closure of a ball B, then the inequalities in (14.3) and (14.4) are true for all $x \in B$.

2. Under the same assumptions as in Theorem 14.2, show that both of the estimates (14.3) and (14.4) can be improved by replacing the integral on their right-hand sides by

$$\int_B \frac{|\nabla f(y)\cdot(x-y)|}{|x-y|^n}\,dy \le \sum_{i=1}^n \int_B \left|\frac{\partial f}{\partial y_i}(y)\right| \frac{|x_i-y_i|}{|x-y|^n}\,dy.$$

3. Derive the L^1, L^1 Poincaré inequality (14.8) directly by the same method used to prove the subrepresentation Theorem 14.2, instead of obtaining it as a corollary of Theorem 14.2.

4. Let B_1 and B_2 be balls, with $B_1 \subset \mathbb{R}^{n_1}$ and $B_2 \subset \mathbb{R}^{n_2}$, $n_1, n_2 \geq 1$. If $f(x_1, x_2) \in C^1(B_1 \times B_2) \cap L^1(B_1 \times B_2)$, show that for every $(x_1, x_2) \in B_1 \times B_2$,

$$\begin{aligned} |f(x_1, x_2) - f_{B_1 \times B_2}| \leq c \int_{B_1 \times B_2} |\nabla_{y_1} f(y_1, y_2)| k_1(x_1, x_2; y_1, y_2)\, dy_1 dy_2 \\ + c \int_{B_1 \times B_2} |\nabla_{y_2} f(y_1, y_2)| k_2(x_1, x_2; y_1, y_2)\, dy_1 dy_2, \end{aligned}$$

where

$$k_i(x_1, x_2; y_1, y_2) = \frac{1}{|B_1||B_2|} \frac{|x_i - y_i|}{\left(\frac{|x_1 - y_1|}{r(B_1)} + \frac{|x_2 - y_2|}{r(B_2)} \right)^{n_1 + n_2}}, \quad i = 1, 2,$$

$f_{B_1 \times B_2} = (|B_1||B_2|)^{-1} \int_{B_1 \times B_2} f$, and c is a constant that is independent of f, x_1, x_2, B_1, and B_2. Note that each integral is at most a multiple of

$$\begin{aligned} \int_{B_1 \times B_2} \big(|\nabla_{y_1} f(y_1, y_2)| r(B_1) + |\nabla_{y_2} f(y_1, y_2)| r(B_2) \big) \\ \times K(x_1, x_2; y_1, y_2)\, dy_1 dy_2 \end{aligned}$$

where

$$K(x_1, x_2; y_1, y_2) = \frac{1}{|B_1||B_2|} \frac{1}{\left(\frac{|x_1 - y_1|}{r(B_1)} + \frac{|x_2 - y_2|}{r(B_2)} \right)^{n_1 + n_2 - 1}}.$$

5. Let Q be a closed cube in $\mathbb{R}^n$, $n > 1$. Show that if $f \in C^1(Q)$, then

$$|f(x) - f_Q| \leq c_n \int_Q \frac{|\nabla f(y)|}{|x-y|^{n-1}}\, dy, \qquad x \in Q,$$

where $f_Q = |Q|^{-1} \int_Q f$ and c_n depends only on n. Also, by choosing Q to be the unit cube centered at the origin and making an affine change of variables, show that if $I = \prod_{i=1}^n [a_i, b_i]$ is an interval in $\mathbb{R}^n$ and $f \in C^1(I)$, then for all $x \in I$,

$$|f(x) - f_I| \leq \frac{c_n}{|I|} \int_I \left(\sum_{i=1}^n (b_i - a_i) \left| \frac{\partial f}{\partial y_i}(y) \right| \right) \frac{dy}{\left(\sum_{i=1}^n \frac{|x_i - y_i|}{b_i - a_i} \right)^{n-1}},$$

where $f_I = |I|^{-1} \int_I f$.

6. Let $I = \prod_{i=1}^{n} [a_i, b_i]$ be an interval in $\mathbb{R}^n$ and suppose that $f \in C^1(I)$. Show that

$$\int_I |f(x) - f_I| \, dx \leq c_n \int_I \sum_{i=1}^{n} (b_i - a_i) \left| \frac{\partial f}{\partial x_i}(x) \right| dx,$$

where $f_I = |I|^{-1} \int_I f$ and c_n is independent of f and I.

7. The proof of Lemma 14.15 was carried out for the ball $B = B(0; r)$ and a point $x = (x, 0, \ldots, 0)$ with $0 \leq x < r$. Show that the proof is similar in the general case.

8. Verify the existence of balls B_x and B_y with the properties described after (14.32), that is, using the notation there and assuming that x, y, r, R, ε satisfy (14.31) and (14.32), show that there are open balls $B_x, B_y \subset B(0;r)$ such that $x \in B_x$, $y \in B_y$, $r(B_x) = r(B_y) = R$, and $B_x \cap B_y \neq \varnothing$. (Denote $d = |x-y|$, $x' = x/|x|$, $y' = y/|y|$, and let $x_d = (|x| - d)x'$ and $y_d = (|y| - d)y'$. Then the balls $B_x = B(x_d; R)$ and $B_y = B(y_d; R)$ have the desired properties. The restriction on ε in (14.31) helps to show that $B_x, B_y \subset B(0;r)$. Also, $(x_d + y_d)/2 \in B_x \cap B_y$. It may be helpful to note that $|x_d - y_d| < d$, for example, by the law of cosines.)

9. Show that a function that satisfies (14.28) can be redefined on a subset of B_0 of measure zero so that (14.29) holds with the same constant C_3 as in (14.28).

10. Prove that Theorems 14.12 and 14.25 remain true if B_0 is replaced by $\mathbb{R}^n$ and f is assumed to be locally integrable on $\mathbb{R}^n$.

11. Show that the function $|x|^{1/2}$ is Hölder continuous of order β on $\mathbb{R}^n$ if and only if $\beta = 1/2$.

12. In the plane $\mathbb{R}^2$, consider the class of rectangles $I((x,y);h)$ with edges parallel to the coordinate axes, center (x,y), x-dimension h, and y-dimension h^2, $h>0$, where (x,y) and h are allowed to vary. Suppose that $f(x,y)$ is a locally integrable function that satisfies

$$\frac{1}{h^3}\iint_{I((x,y);h)} |f(u,v) - f_{I((x,y);h)}|\,du\,dv \le Ch^\beta$$

for some C and β independent of (x,y) and h, with $0<\beta\le 1$. Show that after possible redefinition on a set of measure zero, f satisfies

$$|f(x,y)-f(u,v)| \le c(|x-u|+|y-v|^{1/2})^\beta$$

for all (x,y) and (u,v), where c depends only on C and β. (It may be helpful to consider $\mathbb{R}^2$ endowed with the metric $d((x,y),(u,v)) = |x-u|+|y-v|^{1/2}$ instead of the usual metric.)

13. Verify the dilation property $\delta_\lambda(I_\alpha f) = \lambda^\alpha I_\alpha(\delta_\lambda f)$, where $(\delta_\lambda f)(x) = f(\lambda x)$, $\lambda > 0$, and use it to prove (14.35). Similarly, show that if $0 < \alpha < n$, $q > 0$, and there is a constant c such that

$$\sup_{t>0} t|\{x \in \mathbb{R}^n : |I_\alpha f(x)| > t\}|^{1/q} \leq c\|f\|_1$$

for all f, then $q = n/(n-\alpha)$.

14. Let B be the ball $B(0;1/2)$ in $\mathbb{R}^n$, and let $0 < \alpha < n$ and $0 \le \beta < 1 - (\alpha/n)$. Then the function

$$f(x) = \chi_B(x)[|x|^\alpha |\log|x||^{1-\beta}]^{-1}$$

has compact support and belongs to $L^{n/\alpha}(\mathbb{R}^n)$. Show that there is a positive constant c such that

$$I_\alpha f(x) \ge c \begin{cases} \log\log\dfrac{1}{|x|} & \text{if } \beta = 0, \\ \left(\log\dfrac{1}{|x|}\right)^\beta & \text{if } 0 < \beta < 1 - \dfrac{\alpha}{n} \end{cases}$$

for all x near 0.

In particular,

$$\lim_{x\to 0} I_\alpha f(x) = +\infty,$$

and $I_\alpha f$ is not essentially bounded in any neighborhood of the origin. How are $I_\alpha f$ and $\widetilde{I}_\alpha f$ related?

15. Let $0 < \alpha < n$, $\gamma = 2 - (\alpha/n)$, and $B = B(0; 1/2)$. Show that the function

$$f(x) = \chi_B(x)\{|x|^n |\log |x||^\gamma\}^{-1}$$

belongs to $L^1(\mathbb{R}^n)$ but that $I_\alpha f \notin L^{n/(n-\alpha)}_{\text{loc}}(\mathbb{R}^n)$. (Show that there is a constant $c > 0$ such that $I_\alpha f(x) \geq c|x|^{\alpha-n} |\log |x||^{1-\gamma}$ for all x near the origin.)

16. If $0 < \alpha < n$ and $f \in L^{n/\alpha}(\mathbb{R}^n)$, show that $\widetilde{I}_\alpha f$ is a measurable function. (One way to proceed is to express $\widetilde{I}_\alpha f$ as the limit of a sequence of measurable functions: for example, by (14.43),

$$\widetilde{I}_\alpha f(x) = \lim_{k\to\infty} \int_{|y|<k} f(y) \left\{ \frac{1}{|x-y|^{n-\alpha}} - \frac{\chi(y)}{|y|^{n-\alpha}} \right\} dy$$

for almost every x.)

17. Complete the proof of Theorem 14.44.

18. Let $0 < \alpha < n$ and $f \in L^{n/\alpha}(\mathbb{R}^n)$. The discussion preceding Corollary 14.46 shows that the condition

$$\int_{|y|>1} |f(y)| \frac{1}{|y|^{n-\alpha}}\, dy < \infty$$

is necessary and sufficient for the existence and finiteness a.e. of $I_\alpha f$. Prove that the condition holds if there exists $p \in [1, n/\alpha)$ such that $f \in L^{n/\alpha}(\mathbb{R}^n) \cap L^p(\mathbb{R}^n)$. Consequently, the conclusion of Corollary 14.46 is valid for such f.

19. Let $0 < \beta \leq 1$ and B_0 be a ball in $\mathbb{R}^n$. Suppose that f is a measurable function on B_0 that satisfies

$$\sup_B \frac{1}{|B|^{1+(\beta/n)}} \int_B |f(x) - c(f,B)|\, dx < \infty,$$

where the supremum is taken over all balls $B \subset B_0$ and $c(f,B)$ is a constant depending on f and B. Show that $f \in L^1(B_0)$ and that (14.26) holds. (Argue as in the proof of Lemma 14.49.)

20.

(a) Show that $\log|x| \in \mathrm{BMO}(\mathbb{R}^n)$.

(b) In case $n = 1$, show that the odd function $(\operatorname{sign} x)\log|x|$ does not belong to $\mathrm{BMO}(-\infty, \infty)$. Find an analogue of this fact in case $n > 1$.

(For part (a), it may help to consider two types of balls, those with center $x_B \neq 0$ and radius $r(B) < |x_B|/2$ and those with center x_B and radius $r(B) \geq |x_B|/2$ [e.g., $x_B = 0$], and apply Lemma 14.49. If B is of the first type, use the fact that $\log|x|$ is continuously differentiable in B and choose $c(\log|x|, B) = \log|x_B|$ in the lemma. If B is of the second type, choose $c(\log|x|, B) = \log r(B)$. For part (b), consider intervals centered at the origin.)

21. Show that $|x|^{\lambda} \notin \mathrm{BMO}(\mathbb{R}^n)$, $-\infty < \lambda < \infty$, $\lambda \neq 0$.

Show that $|x|^{\lambda}\chi_{|x|<1}(x) \in \mathrm{BMO}(\mathbb{R}^n)$ if $\lambda \geq 0$ but not if $\lambda < 0$.

22. Give an example of a compactly supported function that belongs to $L^p(\mathbb{R}^n)$ for every p, $0 < p < \infty$, but does not belong to $\mathrm{BMO}(\mathbb{R}^n)$.

23. Derive an analogue of the decomposition Lemma 14.55 for any given $s > 0$ assuming that $f \in L^1(\mathbb{R}^n)$. (Given s and f, adjust the size of the cubes Q in the initial grid, which now covers all of $\mathbb{R}^n$, such that $|Q|^{-1} \int_Q f \leq s$ and all Q have the same edge length.)

24. Let $w(x)$ be measurable and positive a.e. in $\mathbb{R}^n$, and suppose that $\log w \in \mathrm{BMO}(\mathbb{R}^n)$. Show that there are positive constants A_1 and A_2 depending on n, with A_1 also depending on $\|\log w\|_*$, such that

$$\left(\frac{1}{|B|}\int_B w^{A_1}\,dx\right)\left(\frac{1}{|B|}\int_B w^{-A_1}\,dx\right) \le A_2 \quad \text{for all balls } B \subset \mathbb{R}^n.$$

(This can be deduced from Corollary 14.53 by writing

$$\int_B w^{A_1}\,dx = \int_B \exp\{A_1(\log w - (\log w)_B)\}\,dx\ \exp\{A_1(\log w)_B\},$$

together with a similar formula for $\int_B w^{-A_1}\,dx$; note that the function $\log(1/w) = -\log w$ also belongs to $\mathrm{BMO}(\mathbb{R}^n)$.)

25. Let $1 < p < \infty$. A nonnegative function $w(x)$ on $\mathbb{R}^n$ is said to satisfy the A_p condition if $w, w^{-1/(p-1)} \in L^1_{\rm loc}(\mathbb{R}^n)$ and there is a constant C such that for every ball $B \subset \mathbb{R}^n$,

$$\left(\frac{1}{|B|}\int_B w(x)\,dx\right)\left(\frac{1}{|B|}\int_B w(x)^{\frac{1}{p-1}}\,dx\right)^{p-1} \le C.$$

For such w, we will write $w \in A_p$. The opposite inequality with $C = 1$ is a consequence of Hölder's inequality. Note that $0 < w(x) < \infty$ for a.e. x if $w \in A_p$.

(a) Show by direct estimation that $|x|^\gamma \in A_p$ if $-n < \gamma < n(p-1)$. (Consider first the case when $r(B) < |x_B|/2$, where $r(B)$ and x_B denote the radius and center of B.)

(b) Show that $\log w \in \mathrm{BMO}(\mathbb{R}^n)$ if $w \in A_p$. (Consider separately the cases $p = 2$, $p < 2$ and $p > 2$. If $w \in A_2$, show that

$$\sup_B \frac{1}{|B|}\int_B \exp\{|\log w(x) - \log w_B|\}\,dx < \infty, \quad w_B = \frac{1}{|B|}\int_B w(x)\,dx.$$

If $p < 2$ and $w \in A_p$, show that $w \in A_2$. In case $p > 2$, use the fact that if $w \in A_p$ then $w^{-1/(p-1)} \in A_{p'}$, $1/p + 1/p' = 1$.)

26. Let Q_0 be a cube in $\mathbb{R}^n$. We say that $f \in \mathrm{BMO}(Q_0)$ if $f \in L^1(Q_0)$ and

$$\|f\|_{**,Q_0} := \sup_{Q \subset Q_0} \frac{1}{|Q|} \int_Q |f - f_Q| < \infty,$$

where the supremum is taken over all cubes Q in Q_0 that have the same orientation as Q_0. Prove that if $f \in \mathrm{BMO}(Q_0)$, then there are positive constants c_1 and c_2 such that for all such Q and all $\lambda > 0$, (14.54) holds with $\|f\|_{**,Q_0}$ in place of $\|f\|_{**}$.

27. Let $\widetilde{f}$ be the periodic conjugate function of f as defined in Chapter 12:

$$\widetilde{f}(x) = -\frac{1}{\pi}\,\text{p. v.}\int_{|t|\le\pi} f(x-t)\frac{dt}{2\tan\frac{1}{2}t}.$$

(a) If $f \in L^\infty[-\pi,\pi]$, show that $\widetilde{f}$ belongs to the class BMO$[-\pi,\pi]$ defined in Exercise 26 and that

$$\|\widetilde{f}\|_{**,[-\pi,\pi]} \le c\|f\|_{L^\infty[-\pi,\pi]}$$

with c independent of f.

(b) Deduce from part (a) the conclusion of Exercise 20 in Chapter 12.

Let $f \in L^\infty(\mathbb{R}^1)$ and have compact support. Show that the Hilbert transform Hf of f belongs to BMO$(\mathbb{R}^1)$, and derive an analogue for $\|Hf\|_*$ of the estimate in part (a).

(For (a), given a small interval $I \subset [-\pi,\pi]$, let x_I denote the center of I and $2I$ denote the interval concentric with I with length $2|I|$. Decompose f on $(x_I-\pi, x_I+\pi)$ as $f = g+h$ where $g = f\chi_{2I}$. Use Hölder's inequality and Theorem 12.79, applied to the periodic extension of g, to show that $\int_I |\widetilde{g}| \le c|I|\|f\|_{L^\infty[-\pi,\pi]}$. Then estimate $\int_I |\widetilde{h}(x) - \widetilde{h}(x_I)|\,dx$ by using the mean value theorem.)

28. For a measurable function f on $\mathbb{R}^n$ and $0 < \alpha < n$, define the fractional maximal function $M_\alpha f$ of f by

$$M_\alpha f(x) = \sup_{B: x \in B} \frac{1}{r(B)^{n-\alpha}} \int_B |f(y)|\, dy, \qquad x \in \mathbb{R}^n.$$

Show that $M_\alpha f$ is a measurable function on $\mathbb{R}^n$ and that there is a constant c independent of f and x such that $M_\alpha f(x) \le c I_\alpha(|f|)(x)$. As a consequence, the estimates in Theorem 14.37 for $I_\alpha f$ also hold for $M_\alpha f$. Show that the same is true if balls B are replaced by cubes Q in the definition of $M_\alpha f$, with $r(B)$ replaced by the edge length of Q.

29. Let $0 < \alpha < n$ and suppose that f satisfies

$$\int_{\mathbb{R}^n} |f|(1 + \log^+ |f|)\, dx < \infty.$$

Prove that $I_\alpha f \in L^{n/(n-\alpha)}(E)$ for every measurable set $E \subset \mathbb{R}^n$ with $|E| < \infty$, and

$$\int_E |I_\alpha f|^{n/(n-\alpha)} dx \leq C \|f\|_{L^1(\mathbb{R}^n)}^{\alpha/(n-\alpha)} \left(|E| + \int_{\mathbb{R}^n} |f| \log^+ |f|\, dx \right),$$

where C is independent of f and E. (Combine (14.39) in case $p = 1$ with the estimate in Exercise 22 of Chapter 9.)

30. Let $k = 2, 3, \ldots$ and B be a ball in $\mathbb{R}^n$, $n > k$. Show that if $f \in C_0^k(B)$, then

$$|f(x)| \le c \int_B |\nabla^k f(y)| \frac{1}{|x-y|^{n-k}}\, dy, \qquad x \in B,$$

where $|\nabla^k f(y)|$ denotes the sum $\sum_{|\alpha|=k} |D^\alpha f(y)|$, $\alpha = (\alpha_1, \ldots, \alpha_n)$ is a multi-index of nonnegative integers, $|\alpha| = \alpha_1 + \cdots + \alpha_n$, and c is a constant independent of B and f. (In case $k = 2$, apply part (ii) of Corollary 14.6 to f and each component of its gradient, and verify the estimate

$$\int_B \frac{1}{|x-z|^{n-1}} \frac{1}{|y-z|^{n-1}}\, dz \le c_n \frac{1}{|x-y|^{n-2}}, \qquad x, y \in B,$$

when $n > 2$.)

15. Weak Derivatives and Poincaré–Sobolev Estimates

1. Construct an example of a function f that has weak first-order partial derivatives in $\mathbb{R}^n$, $n > 1$, but whose ordinary first-order partial derivatives exist nowhere in $\mathbb{R}^n$.

2. Let c be a real number different from 0. Show that there is no locally integrable function g on $(-1,1)$ such that $\int_{-1}^{1} g\varphi\,dx = c\varphi(0)$ for all $\varphi \in C_0^\infty((-1,1))$.

3. The fact that the Cantor–Lebesgue function f does not have a weak derivative on the interval $(0, 1)$ is an immediate corollary of Theorem 15.6. Verify this fact instead directly from the definition of a weak derivative by choosing test functions that are adapted to the graph of f.

4. Consider the convolution $(\varphi * k)(x)$, $x \in \mathbb{R}^n$, where $\varphi \in \mathrm{Lip}_{\mathrm{loc}}(\mathbb{R}^n)$, $k \in L^1(\mathbb{R}^n)$, and k has compact support. Show that for every $i = 1, \dots, n$, the ordinary derivative $\partial(\varphi * k)/\partial x_i$ exists and is finite everywhere in $\mathbb{R}^n$, and

$$\frac{\partial}{\partial x_i}(\varphi * k)(x) = \int_{\mathbb{R}^n} \frac{\partial \varphi}{\partial y_i}(y) k(x-y)\, dy, \qquad x \in \mathbb{R}^n.$$

Show that the same is true without the assumption that k has compact support if φ satisfies the additional conditions $\varphi, \partial\varphi/\partial y_i \in L^\infty(\mathbb{R}^n)$.

5. Verify Theorem 15.9. (Choose a sequence $\{K_\ell\}_{\ell=1}^{\infty}$ of compact subsets of Ω whose interiors increase to Ω. For each ℓ, use Theorem 15.8 to pick a function $f_\ell \in C_0^\infty(\Omega)$ such that $\|f_\ell - f\|_{L^1(K_\ell)} + \|\nabla f_\ell - \nabla f\|_{L^1(K_\ell)} < 1/\ell$. Show that for every compact $K \subset \Omega$, $\{f_\ell\}$ and $\{\nabla f_\ell\}$ converge in $L^1(K)$ to f and ∇f, respectively. Then use a diagonal process to construct a subsequence $\{\ell_j\}$ of positive integers such that $\{f_{\ell_j}\}$ and $\{\nabla f_{\ell_j}\}$ also converge pointwise a.e. in Ω.)

6. Let Ω be an open set in $\mathbb{R}^n$ and let $f, g \in L^1_{\mathrm{loc}}(\Omega)$. Show that for any $i = 1, \ldots, n$, f has weak partial derivative $\partial f/\partial x_i = g$ if and only if there are functions $\{f_j\}_{j=1}^{\infty} \subset C^\infty(\Omega)$ such that $f_j \to f$ in $L^1(K)$ and $\partial f_j/\partial x_i \to g$ in $L^1(K)$ as $j \to \infty$ for every compact set $K \subset \Omega$.

7. Prove that when $1 \leq p \leq \infty$, $W^{1,p}(\Omega)$ is a Banach space with respect to the norm (15.11) and that it is separable when $1 \leq p < \infty$. Prove that $W^{1,2}(\Omega)$ is a Hilbert space with respect to the inner product (15.12). (To show separability, first note that the repeated Cartesian product of $L^p(\Omega)$ with itself is separable when $1 \leq p < \infty$. Then apply the result in Exercise 23 of Chapter 8.)

8. Complete the proof of Theorem 15.13. Furthermore, given a number $\delta > 0$, show that the approximating functions $\{f_j\}$ can be chosen to have supports in the δ-neighborhood $\{x : d(x, \Omega) < \delta\}$ of Ω, where $d(x, \Omega)$ denotes the distance from x to Ω. In particular, in case Ω is a ball B, if B^* is a larger ball concentric with B, the f_j can be chosen to have supports in B^*.

9. Let $1 \le p \le \infty$ and $f \in W^{1,p}(\mathbb{R}^n)$. For h and x in $\mathbb{R}^n$, define the translated function $(\tau_h f)(x) = f(x+h)$. Show that

$$\|\tau_h f - f\|_{L^p(\mathbb{R}^n)} \le |h| \|\nabla f\|_{L^p(\mathbb{R}^n)}.$$

(Consider first the case when $f \in C_0^\infty(\mathbb{R}^n)$ and derive the estimate $\|\tau_h f - f\|_{L^p(\{|x|<k\})} \le |h| \|\nabla f\|_{L^p(\{|x|<k+|h|\})}$ for $k = 1, 2, \ldots$. If p is finite, the general case follows from approximation and letting $k \to \infty$. If $p = \infty$, apply estimates for L^r when r is finite and let $r \nearrow \infty$.)

10.

(a) If $f \in W^{1,\infty}(\mathbb{R}^n)$, show that

$$|f(x) - f(y)| \le |x - y| \|\nabla f\|_{L^\infty(\mathbb{R}^n)} \quad \text{for a.e. } x, y \in \mathbb{R}^n.$$

In particular, a function f belongs to $W^{1,\infty}(\mathbb{R}^n)$ if and only if there exists $g \in \mathrm{Lip}(\mathbb{R}^n)$ such that $f = g$ a.e. in $\mathbb{R}^n$.

(b) If $f \in L^\infty(\mathbb{R}^n)$, show that $f \in W^{1,\infty}(\mathbb{R}^n)$ if and only if there is a constant C such that $|f(x) - f(y)| \le C|x - y|$ for a.e. $x, y \in \mathbb{R}^n$.

11. (Product Rule) Let $1 \le p \le \infty$ and Ω be an open set in $\mathbb{R}^n$. Prove that if $f, g \in W^{1,p}(\Omega) \cap L^\infty(\Omega)$, then $fg \in W^{1,p}(\Omega) \cap L^\infty(\Omega)$ and

$$\frac{\partial}{\partial x_i}(fg) = f\frac{\partial g}{\partial x_i} + \frac{\partial f}{\partial x_i}g \quad \text{in } \Omega, \quad i = 1, \ldots, n.$$

(Note that both fg and the sum on the right side of the formula belong to $L^p(\Omega)$. It is then enough to check that the sum is the weak derivative $\partial(fg)/\partial x_i$. Use approximation by functions in $C_0^\infty(\mathbb{R}^n)$ if p is finite. If $p = \infty$, use the result for finite p after noting that for every bounded open set $\Omega' \subset \Omega$, both f and g belong to $W^{1,p}(\Omega')$ for all finite p.)

12. (Chain Rule) Let $\phi \in C^1(-\infty, \infty)$ with ϕ' bounded and $\phi(0) = 0$. Let $1 \le p \le \infty$ and Ω be an open set in $\mathbb{R}^n$. If $f \in W^{1,p}(\Omega)$, prove that the composition $(\phi \circ f)(x) = \phi(f(x))$ satisfies $\phi \circ f \in W^{1,p}(\Omega)$ and

$$\frac{\partial}{\partial x_i}(\phi \circ f) = (\phi' \circ f)\frac{\partial f}{\partial x_i}, \qquad i = 1, \ldots, n.$$

(The hypothesis $\phi(0) = 0$ guarantees that $|\phi \circ f| \le \|\phi'\|_\infty |f| \in L^p(\Omega)$. It is then enough to check that the expression on the right side of the formula is the weak partial derivative with respect to x_i of $\phi \circ f$. Approximate f by smooth functions if p is finite. If $p = \infty$, use the result for finite p and bounded open sets in Ω.)

13. Let f have compact support in an open set $\Omega \subset \mathbb{R}^n$ and weak derivative $\partial f/\partial x_i$ in Ω for some i. Show that the set where $\partial f/\partial x_i \neq 0$ is contained in the union of the support of f and a subset of Ω of measure 0. (The result in Exercise 2 of Chapter 7 may be helpful.)

14. Let B be a ball in $\mathbb{R}^n$, $1 < p < n$, and $1/q = 1/p - 1/n$. If f has weak gradient ∇f in B and $|\nabla f| \in L^p(B)$, show that

$$\begin{aligned}\|f\|_{L^q(B)} &\le c_{n,p}\|\nabla f\|_{L^p(B)} + c_n \frac{1}{r(B)}\|f\|_{L^p(B)} \\ &\le c_{n,p}\left(1 + \frac{1}{r(B)}\right)\|f\|_{W^{1,p}(B)},\end{aligned}$$

and if either $f_B = 0$ or f has compact support in B, then

$$\|f\|_{L^q(B)} \le c_{n,p}\|\nabla f\|_{L^p(B)} \le c_{n,p}\|f\|_{W^{1,p}(B)}.$$

15. Let f have weak gradient ∇f in $\mathbb{R}^n$ and suppose that $|\nabla f| \in L^p(\mathbb{R}^n)$ for some p with $1 \le p < n$. Let $1/q = 1/p - 1/n$:

(a) Show that $f \in L^q(\mathbb{R}^n)$ if and only if there is a sequence $\{B_k\}_{k=1}^{\infty}$ of balls increasing to $\mathbb{R}^n$ such that $|f_{B_k}||B_k|^{1/q}$ is bounded in k.

(b) Suppose that $f_B \to 0$ for some sequence of balls $B \nearrow \mathbb{R}^n$. Show that

$$\sup\{|f_B||B|^{1/q}\} < \infty,$$

where the supremum is taken over all balls $B \subset \mathbb{R}^n$.

16. Let $n > 1$ and $B = B(0; 1/2)$ be the ball in $\mathbb{R}^n$ of radius $1/2$ centered at the origin. If $0 < \beta < (n-1)/n$, show that $|\log|x||^\beta$ belongs to $W^{1,n}(B)$ but not to $L^\infty(B)$. Compare Exercise 14 in Chapter 14.

17. Let f and g be finite functions defined on a set $\Omega \subset \mathbb{R}^n$. Prove that the function $h(x) = \max\{f(x), g(x)\}$ (as well as its analogue with max replaced by min) satisfies

$$|h(x) - h(y)| \le \max\{|f(x) - f(y)|, |g(x) - g(y)|\}, \qquad x, y \in \Omega.$$

If Ω is open and $f, g \in \mathrm{Lip}_{\mathrm{loc}}(\Omega)$, deduce that $h \in \mathrm{Lip}_{\mathrm{loc}}(\Omega)$ and that

$$|\nabla h(x)| \le \max\{|\nabla f(x)|, |\nabla g(x)|\} \quad \text{for a.e. } x \in \Omega.$$

In particular, if $-\infty < c < \infty$ and $f \in \mathrm{Lip}_{\mathrm{loc}}(\Omega)$, show that the truncated function

$$f_c(x) = \begin{cases} f(x) & \text{if } f(x) \le c, \\ c & \text{if } f(x) > c \end{cases}$$

satisfies $|\nabla f_c| \le |\nabla f|$ a.e. in Ω. Show also that $|\nabla(|f|)| \le |\nabla f|$ a.e. in Ω.

18. Prove the following analogue of Theorem 15.45 when $1 < p < n$ and $1/q = 1/p - 1/n$. If $f \in W^{1,p}(\mathbb{R}^n)$, then

$$\|f\|_{L^q(\mathbb{R}^n)} \le c_{n,p} \prod_{i=1}^{n} \left\| \frac{\partial f}{\partial x_i} \right\|_{L^p(\mathbb{R}^n)}^{1/n}.$$

(In case $f \in C_0^1(\mathbb{R}^n)$, apply (15.46) to the function $F = |f|^{\delta-1} f$ with $\delta = q(n-1)/n$. Note that $1 < \delta < \infty$, and apply Hölder's inequality to the formula $\|\partial F/\partial x_i\|_{L^1(\mathbb{R}^n)} = \|\delta |f|^{\delta-1} \partial f/\partial x_i\|_{L^1(\mathbb{R}^n)}$ with exponents p' and p.)

19.

(a) Show that there exists $f \in C^\infty(\mathbb{R}^n)$ with $|\nabla f| \in L^p(\mathbb{R}^n)$ for every p, $1 \le p \le \infty$, such that (15.39), (15.46), and the first inequality in the conclusion of Theorem 15.29 fail.

(b) Suppose that $1 \le p < n$, $f \in W^{1,p}(\mathbb{R}^n)$, and $\partial f/\partial x_i = 0$ a.e. in $\mathbb{R}^n$ for some i, $i = 1, \ldots, n$. Prove that $f = 0$ a.e. in $\mathbb{R}^n$. (See Exercise 18 in case $1 < p < n$.)

20. Let $n = 1$, (a, b) be an open interval in $\mathbb{R}^1$, possibly of infinite length, and suppose that f has a weak derivative f' in (a, b). Verify the following analogues of Theorems 15.30 and 15.32:

(a) If $f' \in L^1(a, b)$, then $f \in L^\infty(a, b)$ and

$$\|f - f(c)\|_{L^\infty(a,b)} \leq \|f'\|_{L^1(a,b)} \quad \text{for a.e. } c \in (a, b).$$

(b) If $f' \in L^p(a, b)$ for some p with $1 < p \leq \infty$, then

$$|f(x) - f(y)| \leq \|f'\|_{L^p(a,b)}|x - y|^{1/p'} \quad \text{for a.e. } x, y \in (a, b),$$

where $1/p + 1/p' + 1$.

21. Let $f \in W^{1,1}(\mathbb{R}^n)$. Prove that

$$\|f\|_{L^r(\mathbb{R}^n)} \leq \|f\|_{W^{1,1}(\mathbb{R}^n)}, \qquad 1 \leq r \leq \frac{n}{n-1}.$$

22. Verify the following remarkable fact. Let f and g be locally integrable functions on $\mathbb{R}^n$ that satisfy

$$\frac{1}{|B|}\int_B |f - f_B|\,dx \le Cr(B)\frac{1}{|B|}\int_B |g|\,dx$$

for all balls $B \subset \mathbb{R}^n$, with C independent of B. If $1 < p < n$ and $1/q = 1/p - 1/n$, then for every ball B,

$$\left(\frac{1}{|B|}\int_B |f - f_B|^q\,dx\right)^{1/q} \le C'r(B)\left(\frac{1}{|B|}\int_B |g|^p\,dx\right)^{1/p},$$

where C' depends only on n and C. (Recall Theorem 14.12.)

What can be said if $p \ge n$?

23. Let B be a ball in $\mathbb{R}^n$ with $r(B) = 1$ and $n \geq 2$. Let $k = 1, 2, \ldots$ and $f \in C_0^{k+1}(B)$. If $n = 2k$ or $n = 2k + 1$, show that

$$\|f\|_{L^\infty(B)} \leq c_n \|\nabla^{k+1} f\|_{L^2(B)},$$

where the right-hand side denotes the $L^2(B)$ norm of the sum of the absolute value of every partial derivative of f of order $k + 1$, and c_n is a constant independent of B and f. (One way to proceed is to first apply Corollary 14.6 to obtain $\|f\|_{L^\infty(B)} \leq c\|\nabla f\|_{L^r(B)}$ if $r > n$. Then show that $\|\nabla f\|_{L^r(B)} \leq c\|\nabla^2 f\|_{L^{rn/(r+n)}(B)}$, and continue considering higher order derivatives as necessary.)